ESE 3107 Complete Solutions

Solar Thermal Engineering · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the teachers' notes (Khaja sir, Hasan sir) and the textbooks they follow (Duffie & Beckman, Kalogirou, G. N. Tiwari, Cengel)

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 2 h for Part A and 3 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). Most marks here are theory you can learn by heart: collectors (A5), radiation laws (A3), instruments and radiation components (A2), concentrators (B2) and applications (B4). For numericals, the sun-angle calculator covers A1 and the storage-tank and still calculators cover B3 and B4. FPC performance (B1) and radiation networks (A4) are the hardest: leave them for last.

Section A: The sun, radiation heat transfer and collectors

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A5 Solar collectors: classification, flat plate construction, materials, glazing, evacuated tube and heat pipe	Very likely 100%	Easy	26%
A2 Solar radiation: extraterrestrial vs terrestrial, components, clearness index, measuring instruments, tilted surfaces	Very likely 100%	Easy	46%
A3 Radiation laws: blackbody, Planck, Stefan-Boltzmann, Wien, Kirchhoff, gray body, intensity	Very likely 100%	Easy	61%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered
A1 Sun-earth geometry: angles, solar time, sunshine hours, angle of incidence	Very likely 100%	Medium	79%
✔ Pass line: 79% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.0 h)			

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered

A4 View factors and radiation exchange: relations, networks, reradiating surface, shields	Very likely 100%	Hard	100%

Section B: Collector performance and applications

Block 1: very likely and easy · 0.8 h

Topic	Chance	Level	Covered
B4 Other applications: drying, desalination (solar still), cooling, pool vs pond, solar thermal power	Very likely 100%	Easy	32%
B2 Concentrating collectors: definitions, classification, maximum concentration, tracking, materials	Very likely 100%	Easy	51%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered
B3 Solar water heating and storage: thermosiphon, ICS, direct/indirect, heat pump, storage tank numericals	Very likely 100%	Medium	71%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B1 Flat plate collector performance: losses, U_L, F', F_R, critical radiation level, testing, numericals	Very likely 100%	Hard	100%

✔ Pass line: 100% of Section B marks covered, enough to choose 3 of the 4 questions (about 2.6 h)

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2. **Section A: The sun, radiation heat transfer and collectors**: A5 Solar collectors: classification, flat plate construction, materials, glazing, evacuated tube and heat pipe, A2 Solar radiation: extraterrestrial vs terrestrial, components, clearness index, measuring instruments, tilted surfaces, A3 Radiation laws: blackbody, Planck, Stefan-Boltzmann, Wien, Kirchhoff, gray body, intensity, A1 Sun-earth geometry: angles, solar time, sunshine hours, angle of incidence, A4 View factors and radiation exchange: relations, networks, reradiating surface, shields
3. **Section B: Collector performance and applications**: B4 Other applications: drying, desalination (solar still), cooling, pool vs pond, solar thermal power, B2 Concentrating collectors: definitions, classification, maximum concentration, tracking, materials, B3 Solar water heating and storage: thermosiphon, ICS, direct/indirect, heat pump, storage tank numericals, B1 Flat plate collector performance: losses, U_L, F', F_R, critical radiation level, testing, numericals

One-shot notes: ESE 3107 Solar Thermal Engineering

Sun angles (A1)

$\delta = 23.45 \sin[360(284 + n)/365]$; $\omega = 15(ST - 12)$, morning −; $\cos\theta_z = \cos\varphi\cos\delta\cos\omega + \sin\varphi\sin\delta$; $\alpha_s = 90 - \theta_z$; $\cos\gamma_s = (\cos\theta_z\sin\varphi - \sin\delta)/(\sin\theta_z\cos\varphi)$, sign of ω.

$\cos\omega_s = -\tan\varphi\tan\delta$; $N = 2\omega_s/15$ (derive from $\theta_z = 90°$). ST = clock + $4(L_{loc} - 90)$ + E (Bangladesh). Air mass $m = 1/\cos\theta_z$; $G_{sc} = 1367$ W/m^2.

Khulna Jan 18: δ −20.73, N 10.78 h. Khulna Mar 8, 11 AM: E −11.5 min ⇒ 10:47. Lat 45, Mar 5, 3 PM: sunrise 6:27, α 24.4°, γ_s 50.5°W, surface sunrise 7:22. Nirala: θ 21.4°, θ_z 39.0°, R_b 1.20 (use full cosθ formula).

Radiation and instruments (A2)

Extraterrestrial (outside atmosphere, beam only) vs terrestrial (beam + diffuse). $K_T = H/H_o$; diffuse fraction H_d/H falls as K_T rises (Liu-Jordan, Erbs).

$I_T = I_bR_b + I_d(1 + \cos\beta)/2 + I\rho_g(1 - \cos\beta)/2$; $R_b = \cos\theta/\cos\theta_z$; monthly $\bar{H}_T = \bar{R}\bar{H}$.

Pyranometer: thermopile under glass domes, global; shadow band ⇒ diffuse; beam = G − D.
Pyrheliometer: collimator tube on tracker, DNI.
Campbell-Stokes: glass sphere burns card, > 120 W/m^2.

Radiation laws (A3)

Blackbody: absorbs all, max emitter, diffuse. Planck $E_{b\lambda} = C_1/[\lambda^5(e^{C_2/\lambda T} - 1)]$; $E_b = \sigma T^4$; $\lambda_{max}T = 2898$ μm K. Gray: ε independent of λ. Kirchhoff: ε = α (enclosure equilibrium proof).

$E_b = \int\int I\cos\theta\sin\theta\, d\theta d\varphi = \pi I_b$. $d\omega = dA\cos\theta/r^2$.

2023: $E_{b\lambda}$(3 μm, 600 K) = 520, ω = 4.09×10^{-4} sr, Q = 2.74×10^{-4} W.

View factors and networks (A4)

Reciprocity $A_iF_{ij} = A_jF_{ji}$; summation ΣF = 1; superposition; crossed strings $F_{12} = (\Sigma\text{crossed} - \Sigma\text{uncrossed})/2L_1$. $F_{1\text{-}3} = F_{1\to(2,3)} - F_{1\text{-}2}$. 2019: F_{12} 0.075, F_{21} 0.050; 2022: 0.075.

Surface $R = (1 - \varepsilon)/A\varepsilon$, space $R = 1/AF$. Plates $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$ ⇒ 13.9 kW/m^2. Reradiating $J = E_b$. Triangle duct: −28 kW/m (remove). N shields: q/(N + 1).

Collectors (A5)

FPC (< 80 °C), ETC, CPC; trough, Fresnel (1-axis); dish, tower (2-axis). FPC: glazing, absorber (selective α 0.95, ε 0.1), risers + headers, insulation, casing. Greenhouse effect of glass.

ETC: vacuum kills convection/conduction ⇒ U_L 1-2 vs 4-8 ⇒ flatter efficiency line, better hot/cold/windy. Heat pipe: boil low, condense in manifold, drip back; tilt ≥ 25°; freeze-proof, one-way.

FPC performance (B1)

$Q_u = A_cF_R[S - U_L(T_i - T_a)]$; $\eta = F_R(\tau\alpha) - F_RU_L(T_i - T_a)/I$. $I_c = U_L(T_i - T_a)/(\tau\alpha)$. F′ (local fluid), F_R (inlet fluid) = F′F″.

$F' = (1/U_L)/\{W[1/U_L(D + (W - D)F) + 1/C_b + 1/\pi D_ih_{fi}]\}$; $F = \tanh x/x$, $x = m(W - D)/2$. $U_L = U_t + U_b + U_e$; Hollands Nu; $T_{sky} = T_a - 6$.

2020: F 0.961, F′ 0.912, F_R 0.868, Q_u 2.10 kW, η 65.7%. 2021: U_t 3.39 + U_b 0.6 = 4.0. 2019: U_t 5.40, U_L 6.23. 2022 dark test: 12.4.

Concentrators (B2)

$C = A_a/A_r$; acceptance $2\theta_c$; intercept γ. $C_{max} = 1/\sin^2\theta_s$ = 46,747 (3-D), $1/\sin\theta_s$ = 216 (2-D), θ_s = 0.265°: from $T_r \le T_s$.

Tracking: beam only, small acceptance angle. Single-axis (E-W, N-S, polar) vs two-axis. Pros: high T, small loss area. Cons: no diffuse, tracking, cleaning. Reflector: high specular ρ, durable, cheap.

Water heating and storage (B3)

Passive: thermosiphon (tank ≥ 0.3 m above collector), ICS (night loss: selective coat, covers). Active: direct (open loop, differential thermostat), indirect (antifreeze + heat exchanger). Reverse circulation: height, check valve.

$T^+ = T + [Q_u - L - UA(T - T_a)3600]/mc_p$. 2019: 45 → 86.4 °C. 2022: 40 → 116.9 °C (flag: above boiling, pressurised).

Applications (B4)

Dryers: direct (cabinet), indirect (collector + chamber + chimney), PV fans. Still: $q_{ew} = 0.016273h_{cw}(P_w - P_g)$, $P = e^{25.317 - 5144/T}$ Pa; 50/40 ⇒ 0.282 kg/m^{2}h; 60/50 ⇒ 335 W/m^2.

Desiccant wheel + heat wheel + evaporative cooler, regenerated at 60-90 °C. Absorption LiBr-water. Pond: 3 zones, salt gradient stops convection. Tower: heliostats, receiver, molten salt, turbine. Dish-Stirling 5-25 kW.

Definitions to write word for word

Solar constant: solar energy per unit time on a unit area normal to the rays at mean earth-sun distance outside the atmosphere, 1367 W/m^2. **Air mass**: path length of the beam through the atmosphere relative to the vertical path, $1/\cos\theta_z$. **Hour angle**: angular displacement of the sun east or west of the local meridian, 15°/h. **Declination**: angle of the sun at solar noon with the equatorial plane. **Zenith angle**: angle between the vertical and the sun line. **Solar azimuth**: angle from south to the horizontal projection of the sun line. **Sunshine hour**: time with beam irradiance > 120 W/m^2.

Clearness index: ground radiation / extraterrestrial radiation on a horizontal surface. **Diffuse fraction**: diffuse / global. **Geometric factor** R_b: beam on tilted / beam on horizontal. **Overall tilt factor**: total on tilted / total on horizontal.

Radiosity: all radiation leaving a surface (emitted + reflected). **Irradiation**: all radiation arriving. **Reradiating surface**: adiabatic, net radiation zero, $J = E_b$. **View factor**: fraction of radiation leaving i that strikes j directly. **Gray body**: ε independent of wavelength.

Critical radiation level: irradiance at which gain = losses. **F′**: actual gain / gain with absorber at local fluid temperature. **F_R**: actual gain / gain with whole absorber at inlet temperature. **Aperture**: opening receiving the sun. **Acceptance angle**: angular range accepted without moving. **Concentration ratio**: A_a/A_r. **Intercept factor**: fraction of focused beam hitting the receiver.

Sketches to practise (2 minutes each)

1. Sun angles on a horizon circle (θ_z, α_s, γ_s). 2. Pyranometer (domes, thermopile, shadow band) and pyrheliometer (tube on tracker). 3. Planck curves with Wien peaks. 4. Two-surface and three-surface (reradiating) networks. 5. FPC cross-section with labels (G-A-T-H-I-C). 6. Evacuated tube and heat-pipe tube. 7. Thermal network of FPC. 8. Efficiency line with $F_R(\tau\alpha)$ and slope. 9. Trough, Fresnel, dish, tower. 10. Thermosiphon loop with 0.3 m height. 11. Direct and indirect active systems. 12. Indirect dryer. 13. Solar still. 14. Salt-gradient pond. 15. Power tower with salt tanks. 16. Desiccant wheel chain.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
CPC	compound parabolic collector
CSP	concentrated solar power
DNI	direct normal irradiance
ETC	evacuated tube collector
FPC	flat plate collector
GHI	global horizontal irradiance
ICS	integrated collector storage
kWh	kilowatt-hour
LiBr	lithium bromide
LST	local standard time
MJ	megajoule
PV	photovoltaic
sr	steradian
ST	solar time
WMO	World Meteorological Organization

Section A: The sun, radiation heat transfer and collectors

A5 Solar collectors: classification, flat plate construction, materials, glazing, evacuated tube and heat pipe Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q4(a)** [6] What is a solar collector? Classify by temperature range.
- **2023 Q4(b)** [10] Working principle of an FPC (flat plate collector) with schematic.
- **2023 Q4(c)** [10] Materials for components of (i) FPC, (ii) concentrating collector.
- **2023 Q4(d)** [9] "ETCs perform better than FPCs": justify.
- **2023 Q6(b)** [10] Construction of a heat pipe ETC (evacuated tube collector) with diagram.
- **2022 Q4(a)** [10] What is a solar thermal collector? Classify solar collectors.
- **2022 Q4(c)** [10] Components of an FPC with a neat sketch.
- **2022 Q4(d)** [5] Function of glazing materials.
- **2022 Q4(b)** [10] Working principle of an evacuated tube heat-pipe collector.
- **2021 Q4(a)** [13] Construction details of an FPC with schematic.
- **2021 Q5(b)** [6] Compare materials used for glazing and absorber.
- **2021 Q4(b)** [10] Advantages and limitations of ETCs.
- **2021 Q4(c)** [12] Working principle of an ETC with a heat pipe.
- **2020 Q3(b)** [10] Construction details of a typical FPC with schematic.
- **2020 Q4(b)** [8] Short note on glazing material.
- **2020 Q3(c)** [10] Working principle of an evacuated tube collector; advantages.
- **2019 Q4(a)** [9] What is a solar collector? Heat transfer mechanism in a flat plate collector.
- **2019 Q4(c)** [10] What is an ETC? Working principle.
- **2019 Q5(a)** [5] Why is an ETC preferable over an FPC?

★ REMEMBER IT

- Collector families by temperature: **FPC** (stationary, < 100 °C) → **ETC** (stationary, 50–200 °C, vacuum kills convection and conduction loss) → **concentrating** (tracking, 100–2000 °C: CPC (compound parabolic collector), parabolic trough, Fresnel, dish, central tower).
- FPC layers top to bottom: **G-A-R-T-I-C**: Glazing, Air gap, aRbsorber plate (selective black coating), Tubes (risers and headers), Insulation, Casing.
- Heat pipe: liquid boils at the hot end (in the sun), vapour rises to the condenser at the top, gives heat to the manifold water, condenses and runs back by gravity. One-way, freeze-proof, a broken tube does not stop the system.

Before you start: what a solar collector is

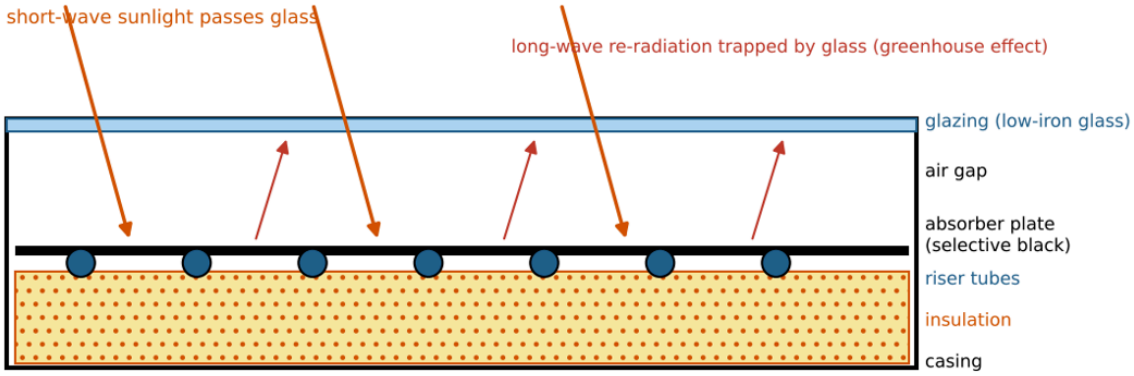
A **solar collector** is a special heat exchanger that absorbs solar radiation, converts it into heat and transfers that heat to a fluid (water, antifreeze, air or oil) flowing through it. Unlike an ordinary heat exchanger, its energy comes from a distant radiant source with a low, variable flux (at most ~1000 W/m^2), so radiation losses matter.

Classification (2022 4a, 2023 4a, 2019 4a)

Type	Collector	Concentration ratio	Typical temperature
Stationary (non-tracking)	Flat plate (FPC)	1	30-80 °C
	Evacuated tube (ETC)	1	50-200 °C
	Compound parabolic (CPC)	1-5	60-240 °C
Single-axis tracking	Linear Fresnel reflector	10-40	60-250 °C
	Parabolic trough	15-45	60-400 °C
Two-axis tracking	Parabolic dish	100-1000	100-500 °C (up to 1500)
	Heliostat field + central receiver	100-1500	150-2000 °C

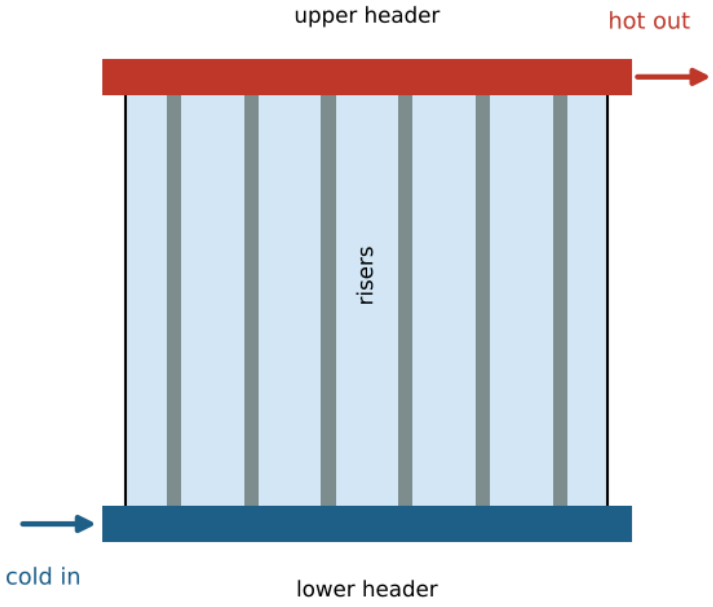
By temperature of application (Hasan sir): low temperature (< 100 °C: domestic hot water, pool heating, drying) — FPC, ETC; medium (100-400 °C: process heat, absorption cooling, small power) — ETC, CPC, trough, Fresnel; high (> 400 °C: electricity, furnaces) — dish, tower. Also by fluid (liquid or air) and glazed/unglazed.

Flat plate collector: construction and working (2020 3b, 2021 4a, 2022 4c, 2023 4b)



Cross-section of a flat plate collector

1. **Glazing (cover)**: one or two sheets of low-iron tempered glass (or plastic); transmits short-wave sunlight ($\tau \approx 0.9$) and is opaque to long-wave re-radiation; also cuts convection loss from the plate.
2. **Absorber plate**: copper, aluminium or steel sheet, painted matte black or with a **selective coating** (black chrome, black nickel, TiNOX: $\alpha \approx 0.95$, $\varepsilon \approx 0.05$-0.15); absorbs sunlight and conducts the heat to the tubes.
3. **Riser tubes and headers (manifolds)**: tubes bonded to the plate carry the fluid; the lower header distributes cold fluid, the upper header collects the hot fluid.
4. **Insulation**: 25-75 mm of glass wool, mineral wool or polyurethane foam at the back and sides to cut conduction loss.
5. **Casing (container)**: aluminium or galvanised steel box with seals and gaskets; holds everything and keeps out dust and moisture.



Fluid path: lower header, risers, upper header

Working: sunlight passes through the glass and is absorbed by the black plate, which heats up. Heat is conducted along the plate (fin) to the tubes and transferred by convection to the fluid, which leaves hotter (by pump or thermosiphon). The plate also loses heat: by radiation and natural convection to the glass and then to the sky and wind (top loss), and by conduction through the insulation (back and edge loss). The glass traps the long-wave radiation emitted by the plate (greenhouse effect). Useful heat = absorbed − losses.

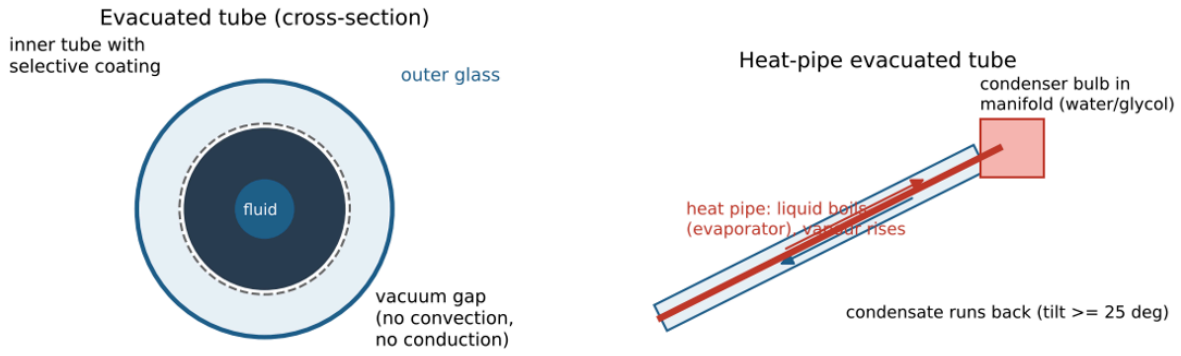
Heat transfer mechanism (2019 4a): radiation: solar irradiation absorbed; long-wave emission from plate to glass and from glass to sky; **conduction**: within the plate to the tubes, through the tube wall and through the insulation; **convection**: from tube wall to the fluid (useful), natural convection in the air gap between plate and glass, and wind convection from the glass and casing (losses).

Glazing and absorber materials (2020 4b, 2022 4d, 2021 5b, 2023 4c)

Component	Required properties	Materials
Glazing	high solar transmittance (0.3-3 μm), opaque to long wave, low iron, resistance to UV, hail, thermal shock; low reflection	low-iron tempered glass (τ 0.85-0.91), anti-reflective glass; polycarbonate, acrylic, Tedlar films (cheaper, lighter, but degrade under UV and are partly transparent to long wave)
Absorber	high α for sunlight, low ε for long wave (selective), high k, corrosion resistance, compatibility with the fluid	copper (k = 385), aluminium, steel; coatings: matte black paint (non-selective), black chrome, black nickel, sputtered cermet (TiNOX)
Insulation	low k, stable up to stagnation temperature (~200 °C), no outgassing	glass wool, mineral wool, polyurethane foam (not at stagnation contact)
Casing	strong, light, weather-proof	anodised aluminium, galvanised steel, fibreglass
Concentrating collector	reflector: high specular reflectance (0.85-0.95), durable; receiver: selective coating, high-temperature stable; cover: glass envelope (often evacuated)	silvered low-iron glass mirrors, polished/anodised aluminium, silvered acrylic films; steel receiver tubes with cermet coating; borosilicate glass envelopes; thermal oil, molten salt, water/steam

Function of glazing: to transmit as much short-wave solar radiation as possible to the absorber while reducing upward heat loss: it blocks the long-wave radiation emitted by the hot plate and suppresses convection by wind; it also protects the absorber from dirt and weather.

Evacuated tube collector (2020 3c, 2019 4c)



Evacuated tube cross-section and heat-pipe evacuated tube

An **ETC** is made of parallel double-walled glass tubes connected to a header. The space between the outer transparent glass tube and the inner tube (coated with a selective absorber) is evacuated (pressure < 10^{-3} Pa). **Working:** sunlight passes through the outer tube and is absorbed by the selective coating; the vacuum suppresses conduction and convection losses (like a thermos flask), so the heat goes to the fluid in or in contact with the inner tube (direct flow U-tube, or via a heat pipe). Because the tubes are round, they intercept the sun well over a wide range of angles.

Advantages and limitations of ETCs; why better than FPCs (2021 4b, 2019 5a, 2023 4d)

Advantages	Limitations
Vacuum eliminates convection and conduction loss: U_L ~1-2 W/m^{2}K vs 4-8 for FPC, so high efficiency at large T − T_a (80-200 °C)	More expensive per m^2; glass tubes are fragile
Works in cold, windy and cloudy weather; useful all year	Lower efficiency than an FPC at low temperature (less absorber area per gross area, gaps between tubes)
Round tubes collect well in the morning and evening (less incidence-angle loss)	Snow does not melt off the front (little heat leaks out)
Tubes can be replaced individually; heat-pipe types are freeze- and scale-resistant	Can overheat/stagnate at high temperature; vacuum may be lost over years
Lighter; can reach medium process-heat temperatures	Heat-pipe types need a minimum tilt (~25°)

Justification "ETCs perform better than FPCs": collector efficiency $\eta = F_R(\tau\alpha) - F_R U_L(T_i - T_a)/I$. Both have similar optical efficiency $F_R(\tau\alpha)$, but the ETC's vacuum makes U_L 3-4 times smaller, so its efficiency line is much flatter: at $(T_i - T_a)/I = 0.08$ m^{2}K/W an FPC may be at 25% while an ETC is still above 55%. This advantage is largest in winter, in wind and for hot-water/process temperatures; only for pool heating near ambient is an FPC as good or better.

Heat-pipe evacuated tube collector (2021 4c, 2022 4b, 2023 6b)

1. **Construction:** each evacuated glass tube contains a flat or curved copper/aluminium absorber fin with a selective coating, bonded to a sealed copper **heat pipe**. The heat pipe is itself

evacuated and holds a small quantity of a working liquid (water with additives, alcohol); its top end is a larger **condenser bulb** that plugs into a dry well in an insulated **manifold (header)** through which water or glycol flows.

2. **Working:** the absorber heats the lower part of the heat pipe (evaporator); the low pressure lets the liquid boil at 25-30 °C; the vapour rises to the condenser bulb, gives up its latent heat to the manifold fluid and condenses; the condensate returns by gravity to the bottom. The cycle repeats as long as the sun shines (a passive, one-way "thermal diode": no reverse flow at night).
3. **Features:** no water inside the glass tube (freeze-proof, no scale, a broken tube does not stop the system), dry connection allows tube replacement without draining, minimum tilt about 25° so the condensate can return.

> **Memory trick:** FPC parts "**G-A-T-H-I-C**": Glazing, Absorber, Tubes, Headers, Insulation, Casing. ETC = "thermos flask for sunlight". Heat pipe = "boil low, condense high, drip back".

A2 Solar radiation: extraterrestrial vs terrestrial, components, clearness index, measuring instruments, tilted surfaces

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q1(d)** [7] Differentiate between terrestrial and extraterrestrial radiation.
- **2023 Q2(a)** [7] Define beam, global radiation and diffuse fraction; daily diffuse fraction vs clearness index.
- **2023 Q3(d)** [8] What is geometric factor? Monthly average daily radiation on tilted surfaces.
- **2023 Q2(d)** [8] Monthly average hourly clearness index; overall tilt factor.
- **2022 Q2(a)** [8] What is diffuse fraction? How is daily diffuse fraction related with daily clearness index?
- **2022 Q1(c)** [10] Working principle of a pyrheliometer.
- **2022 Q2(b)** [8] What is geometric factor? Monthly average daily radiation on tilted surfaces.
- **2021 Q1(c)** [6] Distinguish between extraterrestrial and terrestrial radiation.
- **2021 Q1(e)** [6] What is meant by clearness index and diffuse fraction?
- **2021 Q2(a)** [12] Working principle of pyranometer; how can beam radiation be measured with it?
- **2021 Q2(b)** [8] Incident radiation on a tilted surface; beam radiation on a tilted surface using the geometric factor.
- **2020 Q1(b)** [10] Working principle of pyranometer with schematic diagram.
- **2020 Q2(a)** [8] How to calculate solar radiation on a tilted surface?
- **2019 Q2(a)** [8] Clearness index and diffuse fraction; how is diffuse fraction calculated from clearness index?
- **2019 Q2(b)** [9] Define: geometric factor, beam radiation, extraterrestrial radiation.
- **2019 Q1(b)** [10] What is a pyrheliometer? Working principle.

★ REMEMBER IT

- **Pyra-NO-meter sees the whole dome** (global = beam + diffuse, thermopile under a glass dome; add a shadow band to get diffuse only, then beam = global − diffuse). **Pyr-HELIO-meter stares at Helios** (beam only, long collimating tube that tracks the sun).
- Clearness index $K_T = H/H_o$ (ground / top of atmosphere). Clear day K_T high → diffuse fraction H_d/H low. Cloudy: the opposite.
- Tilted surface: $I_T = I_b R_b + I_d(1 + \cos\beta)/2 + \rho_g I(1 - \cos\beta)/2$: beam × geometric factor + sky it can see + ground it can see.

Before you start: what happens to sunlight on its way down

Outside the atmosphere the sun gives 1367 W/m^2. In the atmosphere some is absorbed (ozone, water vapour, CO_2), some is scattered (air molecules, dust, clouds). What reaches the ground arrives as **beam** (straight from the sun) and **diffuse** (scattered, from the whole sky). Their sum on a horizontal surface is the **global** radiation.

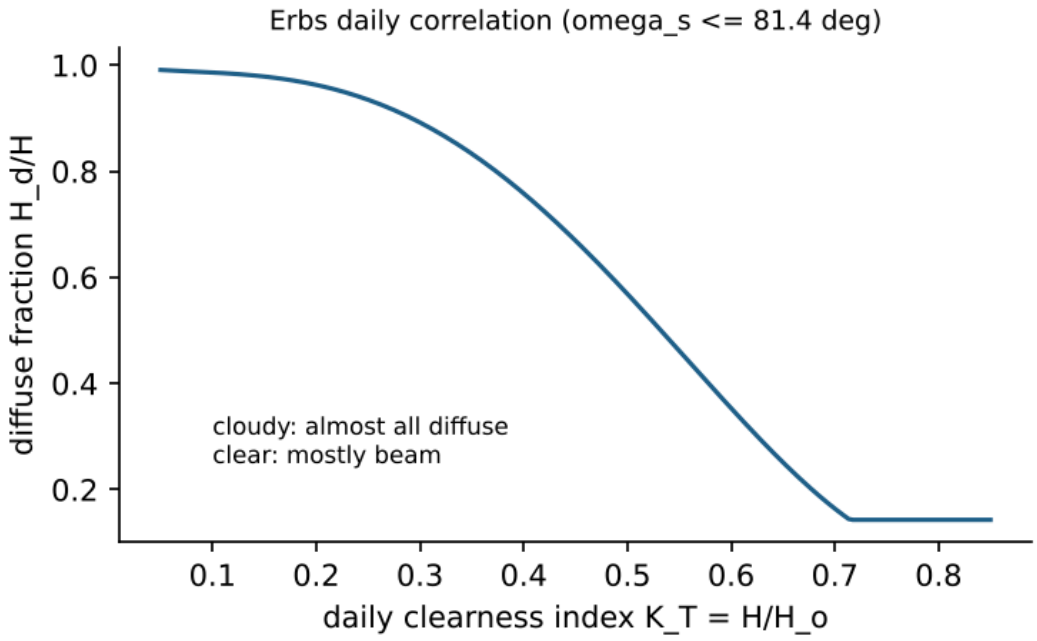
Units: irradiance G or I in W/m^2 (a rate); irradiation H (per day) or I (per hour) in MJ/m^2.

Extraterrestrial vs terrestrial radiation (2021 1c, 2023 1d, 2019 2b)

Extraterrestrial radiation	Terrestrial radiation
Solar radiation outside the earth's atmosphere ($G_{on} = G_{sc}[1 + 0.033\cos(360n/365)]$)	Solar radiation reaching the earth's surface after passing through the atmosphere
Only beam (no scattering); no atmosphere effect	Beam + diffuse; reduced by absorption and scattering
Varies only with earth-sun distance (±3.3%) and geometry; predictable	Varies with weather, clouds, pollution, air mass, altitude; must be measured
Spectrum close to a 5777 K blackbody (0.25-3 μm)	Spectrum has absorption bands (O_3 UV, H_2O and CO_2 infrared)
Used as the reference H_o for the clearness index	Measured with pyranometers and pyrheliometers

Beam radiation: received directly from the sun's disc without scattering. **Diffuse**: received after its direction has been changed by scattering. **Global (total)**: beam + diffuse on a horizontal surface. **Geometric factor** $R_b = \cos\theta/\cos\theta_z$: ratio of beam radiation on a tilted surface to that on a horizontal surface.

Clearness index and diffuse fraction (2021 1e, 2019 2a, 2022 2a, 2023 2a)

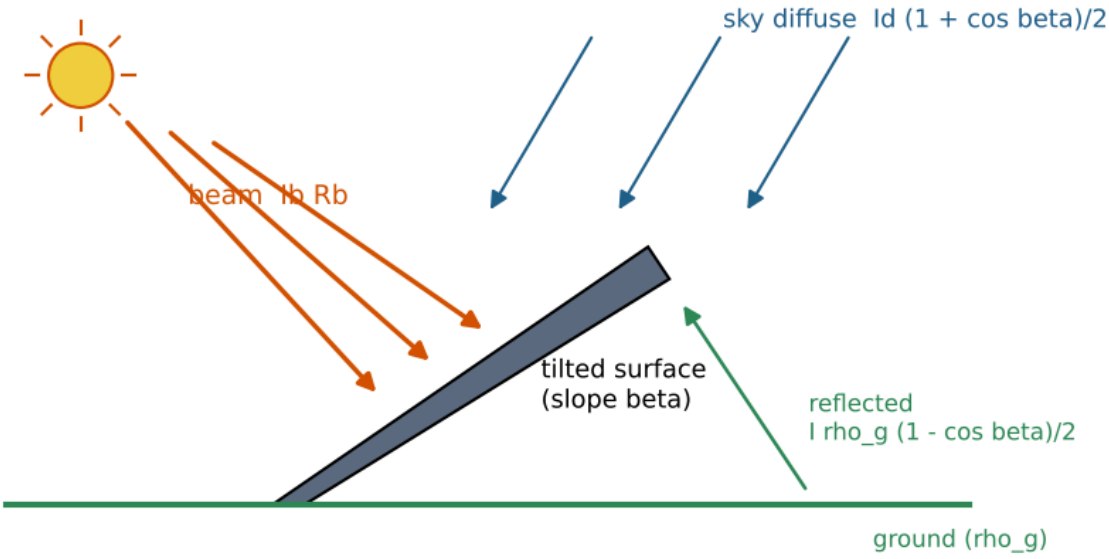


Daily diffuse fraction falls as the clearness index rises

- **Clearness index** $K_T = H/H_o$ (daily), $k_T = I/I_o$ (hourly), $\bar{K}_T = \bar{H}/\bar{H}_o$ (monthly average): fraction of extraterrestrial radiation that reaches the ground. Clear day 0.7-0.75, overcast < 0.3.
- **Diffuse fraction** H_d/H: fraction of the global radiation that is diffuse.
- **Relation:** a clear sky (high K_T) lets the beam through, so the diffuse fraction is small; a cloudy sky (low K_T) scatters almost everything, so the diffuse fraction approaches 1. Empirical correlations: Liu and Jordan $H_d/H = 1.390 - 4.027K_T + 5.531K_T^2 - 3.108K_T^3$; Erbs (for $\omega_s \le 81.4°$) $H_d/H = 1.0 - 0.2727K_T + 2.4495K_T^2 - 11.9514K_T^3 + 9.3879K_T^4$ for $K_T < 0.715$ and 0.143 above. **Procedure:** measure H, compute H_o from geometry, $K_T = H/H_o$, read H_d/H from the correlation, then H_d and $H_b = H - H_d$.

Monthly average hourly clearness index (2023 2d): $\bar{k}_T = \bar{I}/\bar{I}_o$, average over a month of the ratio of hourly global radiation to hourly extraterrestrial radiation for the same hour (I_o from geometry). **Overall tilt factor** $R = I_T/I$ (or H_T/H): ratio of total radiation on the tilted surface to that on a horizontal surface; $R = (1 - I_d/I)R_b + (I_d/I)(1 + \cos\beta)/2 + \rho_g(1 - \cos\beta)/2$.

Radiation on a tilted surface (2020 2a, 2021 2b, 2022 2b, 2023 3d)

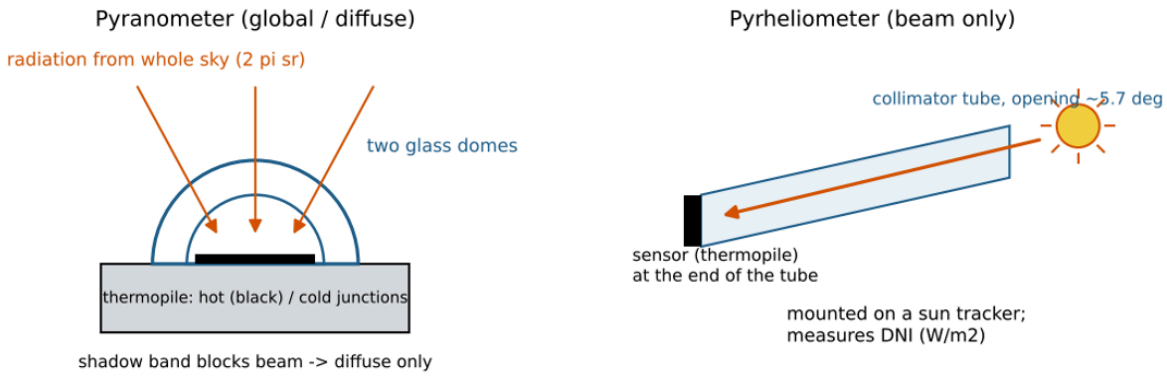


Three components reach a tilted surface

Step (write this)	Why / what it means
$I_T = I_bR_b + I_d(1 + \cos\beta)/2 + I\rho_g(1 - \cos\beta)/2$	Isotropic-sky (Liu-Jordan) model: beam + sky diffuse + ground-reflected.
$R_b = \cos\theta/\cos\theta_z$	Geometric factor converts horizontal beam to the tilted plane (projection of the same beam on two planes).
$(1 + \cos\beta)/2$ = view factor from the surface to the sky; $(1 - \cos\beta)/2$ = view factor to the ground	A horizontal surface sees all sky; a vertical one half sky, half ground.
Beam with the geometric factor: measure/estimate I_b on the horizontal, compute $\cos\theta$ for the surface and $\cos\theta_z$, then $I_{b,T} = I_bR_b$	e.g. R_b = 1.20 at Nirala (A1).

Monthly average daily radiation on a tilted surface (Liu-Jordan, Klein): $\bar{H}_T = \bar{R}\bar{H}$, with $\bar{R} = (1 - \bar{H}_d/\bar{H})\bar{R}_b + (\bar{H}_d/\bar{H})(1 + \cos\beta)/2 + \rho_g(1 - \cos\beta)/2$. Steps: (1) $\bar{H}$ from data, $\bar{H}_o$ for the mean day of the month ⇒ $\bar{K}_T$; (2) $\bar{H}_d/\bar{H}$ from a $\bar{K}_T$ correlation; (3) for a south-facing surface $\bar{R}_b = [\cos(\varphi - \beta)\cos\delta\sin\omega'_s + (\pi/180)\omega'_s\sin(\varphi - \beta)\sin\delta]/[\cos\varphi\cos\delta\sin\omega_s + (\pi/180)\omega_s\sin\varphi\sin\delta]$, $\omega'_s = \min[\omega_s, \cos^{-1}(-\tan(\varphi - \beta)\tan\delta)]$; (4) multiply.

Measuring instruments (2020 1b, 2021 2a, 2019 1b, 2022 1c)



Pyranometer and pyrheliometer

Pyranometer measures **global** (beam + diffuse) radiation on a plane from the whole hemisphere (2π sr). **Working principle:** a black (high-absorptance) sensing surface under one or two hemispherical glass domes is heated by sunlight; a **thermopile** (many thermocouples in series) has its hot junctions under the black surface and cold junctions under a white surface or the body; the temperature difference produces a voltage proportional to the irradiance (about 10 µV per W/m^2). The glass domes pass 0.3-3 µm and protect the sensor from wind and long-wave exchange. Examples: Eppley PSP, Kipp & Zonen CM11.

Measuring beam radiation with a pyranometer (2021 2a): fit a **shadow band (shading ring)** or a tracking shading ball that blocks only the sun's disc, so the pyranometer reads the **diffuse** radiation I_d (corrected for the sky blocked by the band). A second, unshaded pyranometer gives the global I. Then beam on a horizontal surface $I_b = I - I_d$, and direct normal beam $I_{bn} = (I - I_d)/\cos\theta_z$.

Pyrheliometer measures the **beam (direct normal) radiation**. **Working principle:** a long collimating tube with a small opening (aperture angle about 5.7°) is pointed at the sun by a sun-tracker so that only radiation from the sun's disc and a narrow ring around it reaches the black thermopile (or, in an absolute cavity pyrheliometer, a black cavity whose heating is compared with electrical heating) at the bottom of the tube. The output is proportional to the direct normal irradiance (DNI). Examples: Eppley NIP, Ångström compensation pyrheliometer.

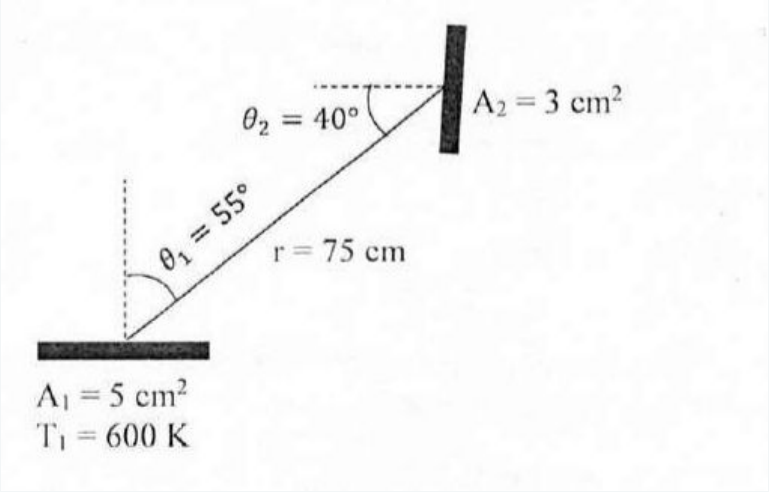
Sunshine recorder (Campbell-Stokes): a 96 mm glass sphere focuses the sun on a card and burns a trace whenever the beam exceeds about 120 W/m^2; the length of the burnt trace gives the sunshine hours. Modern versions use photodiodes.

> **Memory trick: "Pyra-NO-meter sees the whole dome; pyr-HELIO-meter stares at Helios"** (the sun). K_T up ⇒ diffuse fraction down. I_T = beam × R_b + diffuse × $(1 + \cos\beta)/2$ + reflected × $(1 - \cos\beta)/2$.

A3 Radiation laws: blackbody, Planck, Stefan-Boltzmann, Wien, Kirchhoff, gray body, intensity Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q2(b)** [8] What is a blackbody? Show that $E_b = \pi I_b$.
- **2023 Q2(c)** [12] $A_1 = 5\ cm^2$ blackbody at 600 K, $A_2 = 3\ cm^2$ at r = 75 cm, $\theta_1 = 55°$, $\theta_2 = 40°$: spectral emissive power at 3 μm, solid angle, radiation striking A_2.



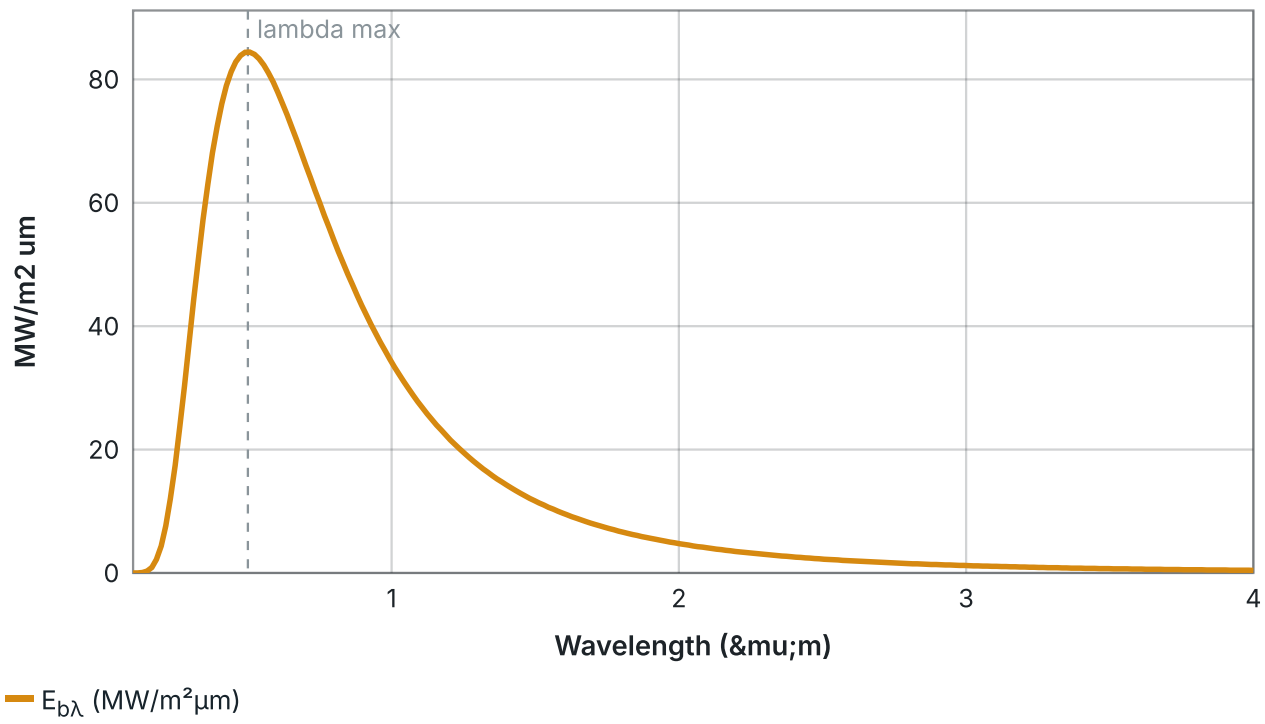
- **2022 Q2(c)** [9] Black body; state and explain the Stefan-Boltzmann law.
- **2022 Q3(c)** [6] Explain Kirchhoff's law of radiation.
- **2022 Q2(d)** [10] Define: emissive power, emissivity, gray body, radiosity, summation rule of view factor.
- **2021 Q2(c)** [9] Black body; state and explain the Stefan-Boltzmann law.
- **2021 Q2(d)** [6] State Kirchhoff's law.
- **2020 Q2(b)** [12] Features of a blackbody; show that $E_b = \pi I_b$.
- **2019 Q2(c)** [10] Explain the Stefan-Boltzmann law and Wien's displacement law.
- **2019 Q2(d)** [8] What is a gray body? Explain Kirchhoff's law.
- **2019 Q3(b)** [9] Define: radiosity, reradiating surface, irradiation.

UNDERSTAND **Read this to understand; you do not need to write it**

Every hot body glows. A **blackbody** is the perfect absorber and emitter. **Planck** gives its full spectrum; **Wien** says the peak moves to shorter wavelengths as it gets hotter ($\lambda_{max}T = 2897.8$ μm K); **Stefan-Boltzmann** gives the total, $E_b = \sigma T^4$. This is why a collector works: sunlight (5800 K) arrives at about 0.5 μm and passes through glass; the hot plate (350 K) emits at about 8 μm and glass blocks it.

INTERACTIVE **Planck's blackbody curve and Wien's peak**

Move the sliders and watch the curve change.



Wien peak λ_{max} = 2897.8/T: **0.500** μm · Total $E_b = \sigma T^4$: **64.16** MW/m²

Visible light is 0.38–0.78 μm. Drag T down to 1000 K: the peak slides far into the infrared and the curve shrinks (T^4).

Before you start: thermal radiation

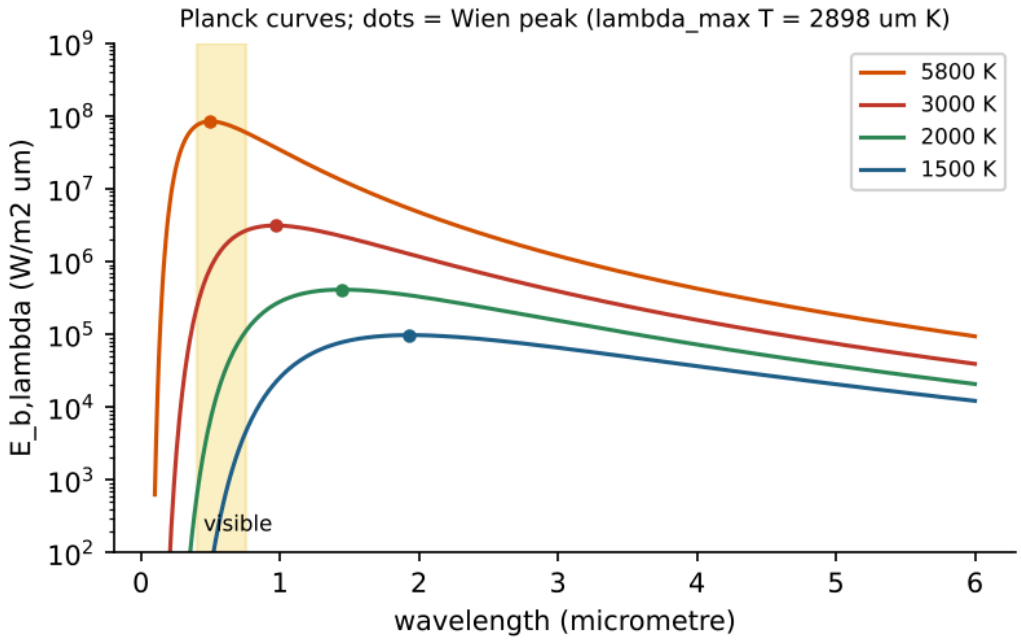
Every body above 0 K emits electromagnetic radiation because of its temperature. Hot bodies emit more and at shorter wavelengths: the sun (5800 K) mostly 0.3-3 μm (short-wave), a collector plate (350 K) mostly 5-50 μm (long-wave). This difference is what makes glazing and selective surfaces work.

Absorptivity α + reflectivity ρ + transmissivity τ = 1 for any surface; opaque surface τ = 0.

Blackbody and its features (2020 2b, 2021 2c, 2022 2c, 2023 2b)

A **blackbody** is a perfect emitter and absorber of radiation. Features: (1) it absorbs **all** incident radiation, of every wavelength and direction (α = 1); (2) at a given temperature and wavelength no surface can emit more energy than a blackbody; (3) it is a **diffuse** emitter: the emitted intensity is the same in all directions; (4) its emission depends only on temperature. A small hole in a large isothermal cavity is the practical approximation.

Radiation laws (2019 2c, 2021 2c, 2022 2c)



Blackbody spectral emissive power

- **Planck's law**: $E_{b\lambda}(\lambda, T) = C_1/[\lambda^5(e^{C_2/\lambda T} - 1)]$, $C_1 = 3.742\times10^8$ W μm^4/m^2, $C_2 = 1.439\times10^4$ μm K: emitted energy per unit wavelength.
- **Stefan-Boltzmann law**: the total energy emitted by a blackbody per unit area and time is proportional to the fourth power of its absolute temperature: $\mathbf{E_b = \sigma T^4}$, $\sigma = 5.67\times10^{-8}$ W/m^{2}K^4. It is the area under the Planck curve: $E_b = \int_0^\infty E_{b\lambda} d\lambda$. A real surface emits $E = \varepsilon\sigma T^4$. Example: doubling T (K) multiplies the emission by 16.
- **Wien's displacement law**: the wavelength of maximum emission is inversely proportional to the temperature: $\mathbf{\lambda_{max} T = 2897.8\ \mu m\ K}$. Sun 5800 K ⇒ 0.50 μm (visible); plate at 350 K ⇒ 8.3 μm (infrared). Obtained by setting $dE_{b\lambda}/d\lambda = 0$.

Gray body and Kirchhoff's law (2019 2d, 2021 2d, 2022 3c)

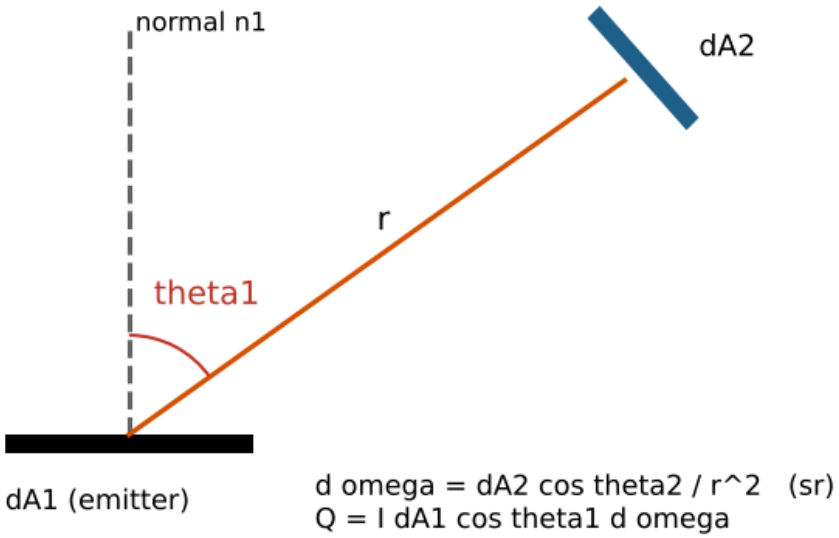
Emissive power E: radiation energy emitted per unit area per unit time (W/m^2). **Emissivity** $\varepsilon = E/E_b$ at the same temperature ($0 \le \varepsilon \le 1$). A **gray body** has emissivity (and absorptivity) independent of wavelength; a **diffuse** surface has properties independent of direction. **Radiosity** J: total radiation leaving a surface (emitted + reflected) per unit area, $J = \varepsilon E_b + \rho G$. **Irradiation** G: total radiation incident on a surface per unit area. **Reradiating surface**: an adiabatic surface with zero net radiation (it re-emits all it receives), so $J = E_b$.

Step (write this)	Why / what it means
`Small body (area A, ε, α) inside a large isothermal enclosure at T, in thermal equilibrium`	Kirchhoff's thought experiment.
`Irradiation on the body G = σT⁴ (blackbody cavity radiation)`	Inside an isothermal enclosure the radiation is blackbody radiation.
`Equilibrium: absorbed = emitted ⇒ AαGσT⁴ = AεσT⁴`	No net heat flow at equal temperatures.
`ε(T) = α(T)`	Kirchhoff's law: at the same temperature, total hemispherical emissivity equals absorptivity;

	spectral form $\varepsilon_\lambda = \alpha_\lambda$ holds always.

Consequence for collectors: a good absorber is a good emitter at the same wavelength. A **selective surface** exploits the wavelength separation: high α in 0.3-2.5 µm (sunlight), low ε beyond 3 µm (its own emission), e.g. black chrome α = 0.95, ε = 0.1.

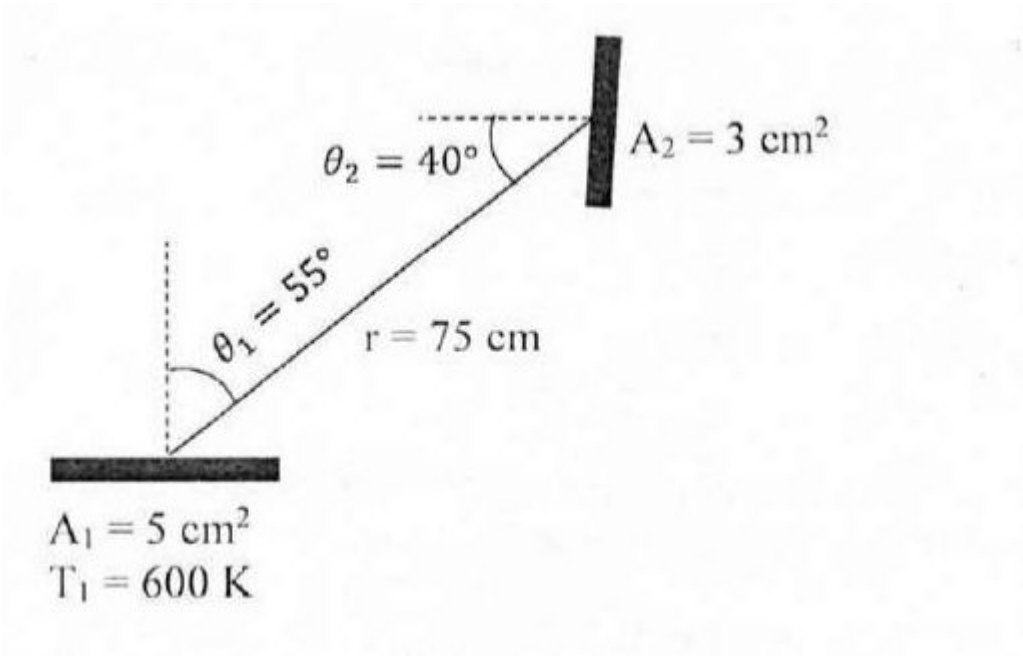
Emissive power of a blackbody = π × intensity (2020 2b, 2023 2b)



Intensity and solid angle

Step (write this)	Why / what it means
Intensity I: energy emitted per unit time, per unit area *normal to the direction*, per unit solid angle (W/m^2sr)	Definition.
Energy from dA through solid angle dω in direction θ: $dE = I \cos\theta \, d\omega$	The projected area is dA cosθ.
On a hemisphere: $d\omega = \sin\theta \, d\theta \, d\varphi$	Solid angle element in spherical coordinates.
$E = \int_0^{2\pi}\int_0^{\pi/2} I \cos\theta \sin\theta \, d\theta \, d\varphi$	Integrate over the whole hemisphere above the surface.
Blackbody (diffuse): I = I_b constant ⇒ $E_b = I_b \times 2\pi \times [\sin^2\theta/2]_0^{\pi/2} = I_b \times 2\pi \times ½$	
$E_b = \pi I_b$	The factor is π (not 2π) because of the cosθ projection.

Numerical: radiation between two small surfaces (2023 2c)



Question figure (2023 paper)

Step (write this)	Why / what it means
(i) $E_{b\lambda} = 3.742\times10^8/[3^5(e^{14{,}390/(3\times600)} - 1)] = 3.742\times10^8/[243\times(e^{7.994} - 1)]$	Planck's law at λ = 3 μm, T = 600 K.
$E_{b\lambda} = 520\ W/m^2\mu m$	
(ii) **$\omega_{2\text{-}1} = A_2\cos\theta_2/r^2 = 3\times10^{-4}\times\cos40°/0.75^2 = 4.09\times10^{-4}\ sr$**	θ_2 = angle between the normal of A_2 and the line joining the surfaces.
$I_1 = E_b/\pi = \sigma T^4/\pi = 5.67\times10^{-8}\times600^4/\pi = 2339\ W/m^2sr$	Blackbody intensity.
(iii) **$Q_{1\to2} = I_1A_1\cos\theta_1\omega_{2\text{-}1} = 2339\times5\times10^{-4}\times\cos55°\times4.09\times10^{-4} = 2.74\times10^{-4}\ W$**	Same method as CengelCengel Example 12-?style.rsquo;s solid-angle examples.

Memory trick: "Black = takes all, gives most". σT^4, $\lambda_{max}T = 2898$, $\varepsilon = \alpha$, $E = \pi I$ (cos-weighted hemisphere).

A1 Sun-earth geometry: angles, solar time, sunshine hours, angle of incidence Pass set Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q1(a)** [10] Define: solar constant, azimuth angle, irradiation, air mass.
- **2023 Q1(c)** [6] What is sunshine hour? How is it related to declination? Explain.
- **2023 Q1(b)** [12] Angle of incidence on a surface at Nirala (22.8°N), 11:00 solar time, February 15, tilted 25°, 8° west of south; geometric factor.
- **2022 Q1(a)** [15] Define: air mass, hour angle, solar azimuth angle, beam radiation, zenith angle.
- **2022 Q1(b)** [10] Derive $N = (2/15)\cos^{-1}(-\tan\varphi \tan\delta)$.
- **2021 Q1(b)** [9] Define with schematic diagram: zenith angle, solar altitude angle, solar azimuth angle.
- **2021 Q1(a)** [7] What is sunshine hour? How is it related to declination?
- **2021 Q1(d)** [7] Sunshine hours for Khulna (22.80°N, 89.24°E) on January 18.
- **2020 Q1(a)** [9] Define: sunshine hour, clearness index, hour angle.
- **2020 Q1(c)** [11] Solar time corresponding to 11 AM on March 8 at Khulna (89.54°E).
- **2019 Q1(a)** [15] Define: solar constant, air mass, solar time, hour angle, surface azimuth angle.
- **2019 Q1(c)** [10] 60° sloped surface facing 25° west of south, 3:00 PM solar time, March 5, latitude 45°: sunrise, solar altitude and azimuth; sunrise and sunset on the surface.

LEARN IT FROM ZERO

Sun angles, from zero

Every solar numerical asks one thing: **where is the sun in my sky right now, and how squarely does it hit my surface?** Three numbers decide it.

- **Latitude φ**: where you are (Khulna 22.8°N).
- **Declination δ**: which season. The earth's axis is tilted 23.45°, so the sun's noon position moves north and south through the year: $\delta = 23.45 \sin[360(284 + n)/365]$, n = day of the year. +23.45° on 21 June, −23.45° on 21 December, 0 at the equinoxes.
- **Hour angle ω**: what time it is. The earth turns 15° per hour, so $\omega = 15$(solar time − 12): − in the morning, + in the afternoon, 0 at solar noon.

From these: zenith angle $\cos\theta_z = \cos\varphi\cos\delta\cos\omega + \sin\varphi\sin\delta$; sunset hour angle $\cos\omega_s = -\tan\varphi\tan\delta$; day length $N = 2\omega_s/15$. For a tilted surface the long cosθ equation adds tilt β and surface azimuth γ (0 = facing south, west positive). **$R_b = \cos\theta/\cos\theta_z$** tells how much more beam radiation the tilted surface gets than flat ground.

Solar time is not clock time: correct for your longitude (4 minutes per degree away from the 90°E Bangladesh meridian) and for the equation of time E (the earth's orbit is elliptical).

CALCULATOR **Sun angles for any day, place, time and surface (2019 1(c), 2021 1(d), 2023 1(b))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Declination δ: $\delta = 23.45 \sin[360(284 + n)/365] = 23.45 \times \sin(360 \times (284 + \mathbf{46}) / 365) =$ **-13.29 °**
2. Hour angle ω: $\omega = 15(t - 12) = 15 \times (\mathbf{11} - 12) =$ **-15.0 °**
3. Sunset hour angle ω_s: $\cos\omega_s = -\tan\varphi \tan\delta = \cos^{-1}(-\tan(\mathbf{22.8}) \times \tan(\mathbf{(-13.29)})) =$ **84.30 °**
4. Day length N: $N = 2\omega_s/15 = 2 \times \mathbf{84.3} / 15 =$ **11.24** h
5. Sunrise (solar time): 12 - **84.3** / 15 = **6.38** h
6. $\cos\theta_z$: $\cos(\mathbf{22.8}) \times \cos(\mathbf{(-13.29)}) \times \cos(\mathbf{(-15)}) + \sin(\mathbf{22.8}) \times \sin(\mathbf{(-13.29)}) =$ **0.7775**
7. Solar altitude: $\alpha_s = 90° - \theta_z = 90 - \cos^{-1}(\mathbf{0.7775}) =$ **51.0 °**
8. cosθ on the tilted surface: $\sin(\mathbf{(-13.29)}) \times \sin(\mathbf{22.8}) \times \cos(\mathbf{25}) - \sin(\mathbf{(-13.29)}) \times \cos(\mathbf{22.8}) \times \sin(\mathbf{25}) \times \cos(\mathbf{8}) + \cos(\mathbf{(-13.29)}) \times \cos(\mathbf{22.8}) \times \cos(\mathbf{25}) \times \cos(\mathbf{(-15)}) + \cos(\mathbf{(-13.29)}) \times \sin(\mathbf{22.8}) \times \sin(\mathbf{25}) \times \cos(\mathbf{8}) \times \cos(\mathbf{(-15)}) + \cos(\mathbf{(-13.29)}) \times \sin(\mathbf{25}) \times \sin(\mathbf{8}) \times \sin(\mathbf{(-15)}) =$ **0.9310**
9. Angle of incidence θ: $\cos^{-1}(\mathbf{0.931}) =$ **21.4 °**
10. Geometric factor R_b: $R_b = \cos\theta/\cos\theta_z = \mathbf{0.931} / \mathbf{0.7775} =$ **1.197**

Day numbers: Jan 1 = 1, Feb 1 = 32, Mar 1 = 60, Apr 1 = 91, May 1 = 121, Jun 1 = 152, Jul 1 = 182, Aug 1 = 213, Sep 1 = 244, Oct 1 = 274, Nov 1 = 305, Dec 1 = 335. 1.5 h = 1 h 30 min.

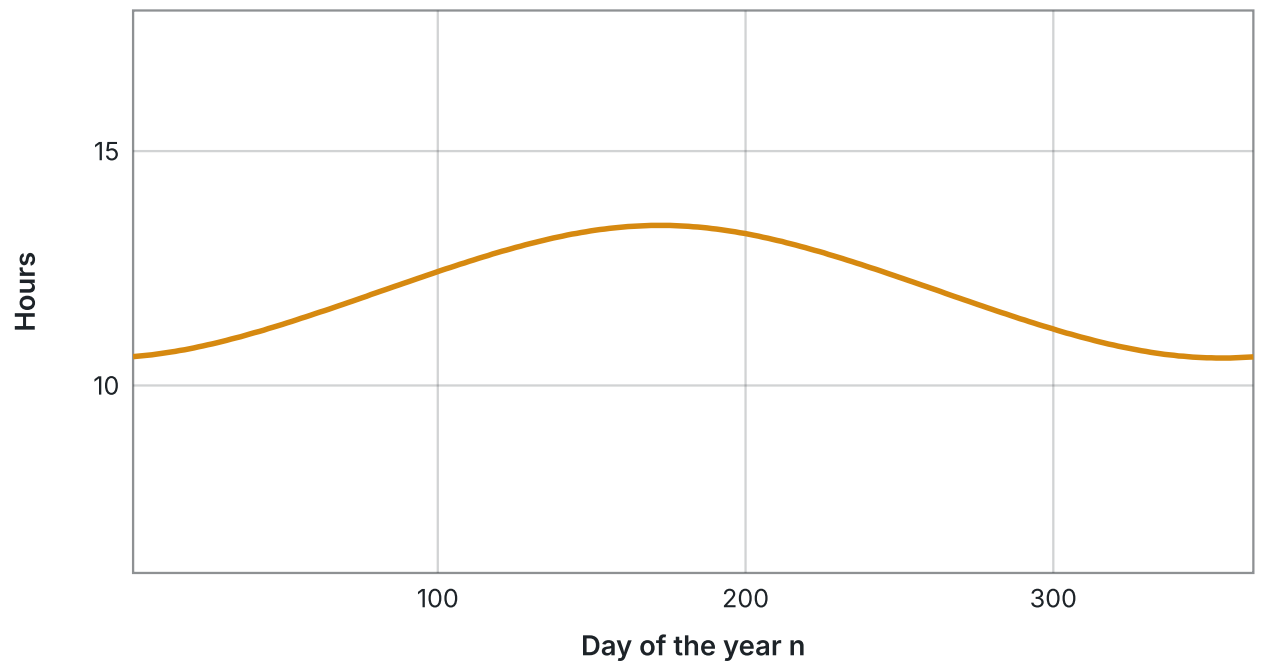
CALCULATOR Clock time to solar time in Bangladesh (2020 1(c))

Change any number; every step is recalculated and shown with the numbers put in.

1. B = (n − 1)360/365: (**67** - 1) × 360 / 365 = **65.10 °**
2. Equation of time E: 229.2 × (0.000075 + 0.001868 × cos(**65.1**) - 0.032077 × sin(**65.1**) - 0.014615 × cos(2 × **65.1**) - 0.04089 × sin(2 × **65.1**)) = **-11.47** min
3. Longitude correction $4(L - L_{st})$: 4 × (**89.54** - **90**) = **-1.84** min
4. Solar time: solar time = clock + $[4(L - L_{st}) + E]/60$ = **11** + (**(-1.84)** + **(-11.47)**) / 60 = **10.778** h
5. Minutes past the hour: (**10.78** - floor(**10.78**)) × 60 = **46.7** min

INTERACTIVE Day length through the year

Move the sliders and watch the curve change.



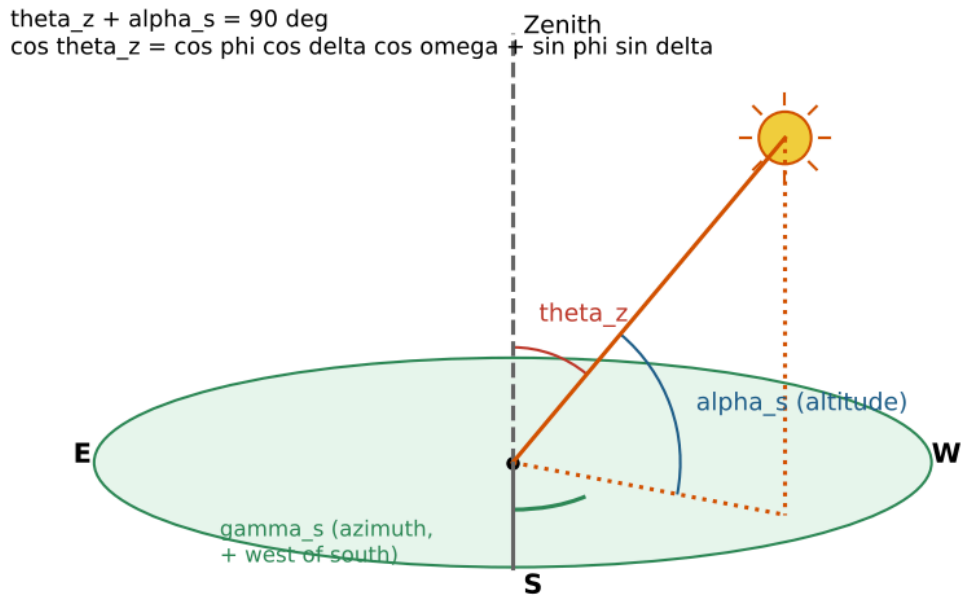
Longest day (n = 172): **13.41** h · Shortest day (n = 355): **10.59** h

At the equator every day is 12 h. The further from the equator, the bigger the summer-winter swing.

Before you start: where the sun is

To know how much sunlight a surface receives, we must know the direction of the sun. Five angles do this: the **latitude** φ of the place, the **declination** δ (which season), the **hour angle** ω (what time of day), and for a tilted surface its **slope** β and **azimuth** γ.

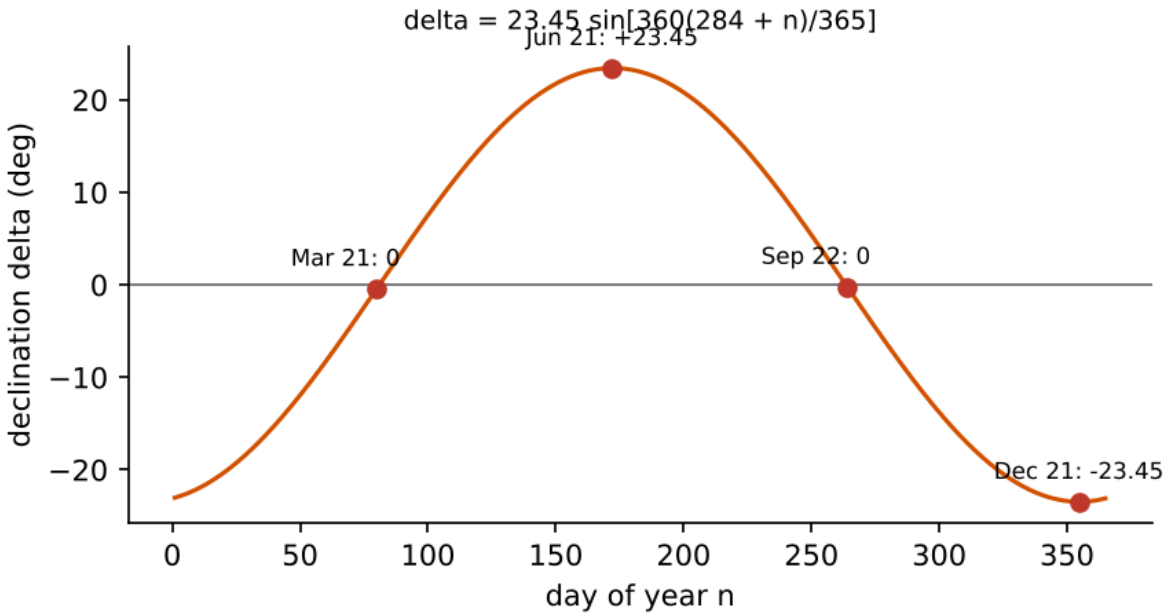
Sign rules (Duffie & Beckman): north latitude +; ω = 0 at solar noon, morning −, afternoon +, 15° per hour; azimuth 0 facing south, east −, west +.



Zenith angle, solar altitude angle and solar azimuth angle

Definitions (2019 1a, 2020 1a, 2021 1b, 2022 1a, 2023 1a)

Term	Meaning
Solar constant G_{sc}	energy from the sun per unit time received on a unit area perpendicular to the rays at the mean earth-sun distance, outside the atmosphere: **1367 W/m^2** (WMO; some books 1361)
Air mass m	ratio of the mass of atmosphere the beam passes through to the mass it would pass through if the sun were overhead: m = 1/cosθ_z (for θ_z < 70°). m = 0 outside the atmosphere, m = 1 sun at zenith, m = 1.5 is the standard test condition
Zenith angle θ_z	angle between the vertical (zenith) and the line to the sun; cosθ_z = cosφcosδcosω + sinφsinδ
Solar altitude angle α_s	angle between the horizontal and the line to the sun; α_s = 90° − θ_z
Solar azimuth angle γ_s	angle between due south and the projection of the sun's rays on the horizontal plane; east of south negative, west positive
Surface azimuth angle γ	angle between due south and the projection of the normal to the surface on the horizontal plane (same sign rule)
Hour angle ω	angular displacement of the sun east or west of the local meridian because of the earth's rotation at 15°/h: ω = 15(ST − 12); morning negative
Declination δ	angle between the sun-earth line and the equatorial plane at solar noon: δ = 23.45 sin[360(284 + n)/365]; +23.45° on June 21, −23.45° on December 21
Solar time ST	time based on the apparent motion of the sun: solar noon is when the sun crosses the local meridian; differs from clock time by a longitude correction and the equation of time
Sunshine hour	length of time in a day during which direct (beam) solar irradiance exceeds 120 W/m^2 (WMO); measured with a sunshine recorder. The maximum possible value is the day length N
Beam radiation	solar radiation received from the sun without being scattered by the atmosphere (direct radiation)
Irradiation	incident energy per unit area on a surface, found by integrating irradiance over a time (J/m^2): H per day, I per hour
Clearness index K_T	ratio of radiation on a horizontal surface at the ground to the extraterrestrial radiation on a horizontal surface for the same period: K_T = H/H_o



Declination through the year

Sunshine hours (day length) and its relation to declination (2021 1a, 2023 1c, 2022 1b)

Step (write this)	Why / what it means
At sunrise and sunset the sun is on the horizon: $\theta_z = 90°$, $\cos\theta_z = 0$	Definition of sunrise/sunset.
$0 = \cos\phi\cos\delta\cos\omega_s + \sin\phi\sin\delta$	Zenith-angle equation with $\omega = \omega_s$.
$\cos\omega_s = -\sin\phi\sin\delta/(\cos\phi\cos\delta) = \mathbf{-\tan\phi\ \tan\delta}$	Sunset hour angle.
The sun moves 15° per hour and the day runs from $-\omega_s$ to $+\omega_s$: total angle $2\omega_s$	
$\mathbf{N = (2/15)\cos^{-1}(-\tan\phi\ \tan\delta)}$ hours	Maximum possible sunshine hours (day length).

Relation to declination: for a northern-hemisphere place ($\varphi > 0$), when $\delta > 0$ (summer) $-\tan\varphi\tan\delta < 0$, $\omega_s > 90°$ and the day is longer than 12 h; when $\delta < 0$ (winter) the day is shorter; at the equinoxes ($\delta = 0$) N = 12 h everywhere. At the equator ($\varphi = 0$) N = 12 h all year.

Numerical 1: sunshine hours at Khulna on January 18 (2021 1d)

Step (write this)	Why / what it means
$n = 18$; $\delta = 23.45\ \sin[360(284 + 18)/365] = 23.45\ \sin(297.9°) = -20.73°$	Day number counted from January 1.
$\cos\omega_s = -\tan(22.80°)\tan(-20.73°) = 0.1591 \Rightarrow \omega_s = 80.85°$	
$\mathbf{N = 2 \times 80.85/15 = 10.78\ h \approx 10\ h\ 47\ min}$	Winter: shorter than 12 h. Sunrise at 6:37 and sunset at 17:23 solar time. Longitude is not needed.

Numerical 2: solar time at Khulna, 11 AM on March 8 (2020 1c)

Step (write this)	Why / what it means
Solar time = standard time + $4(L_{loc} - L_{st}) + E$	For east longitudes; 4 minutes per degree. Bangladesh standard time is GMT + 6 h, so L_{st} = 90°E.

$n = 31 + 28 + 8 = 67$; $B = (n - 1)360/365 = 65.1°$	
$E = 229.2(0.000075 + 0.001868\cos B - 0.032077\sin B - 0.014615\cos 2B - 0.04089\sin 2B) = -11.5$ min	Equation of time: the sun is "late" in early March.
Longitude correction $= 4(89.54 - 90) = -1.8$ min	Khulna is slightly west of the 90°E meridian.
Solar time = 11:00 - 1.8 - 11.5 = 10:46.7 ≈ 10:47 AM	

Numerical 3: sloped surface at latitude 45° on March 5 (2019 1c)

Step (write this)	Why / what it means
$n = 64$; $\delta = 23.45 \sin[360(284 + 64)/365] = -6.76°$; at 3:00 PM $\omega = 15(15 - 12) = +45°$	
Sunrise (horizontal): $\cos\omega_s = -\tan 45° \tan(-6.76°) = 0.1186 \Rightarrow \omega_s = 83.19° = 5.55$ h ⇒ **sunrise 6:27 AM, sunset 5:33 PM (solar time)**	
$\cos\theta_z = \cos 45 \cos(-6.76) \cos 45 + \sin 45 \sin(-6.76) = 0.4132 \Rightarrow \theta_z = 65.6° \Rightarrow$ **solar altitude $\alpha_s = 24.4°$**	
$\cos\gamma_s = (\cos\theta_z \sin\varphi - \sin\delta)/(\sin\theta_z \cos\varphi) = 0.637 \Rightarrow \gamma_s = +50.5°$ **(west of south)**	Sign of γ_s = sign of ω (afternoon: west).
Angle of incidence on the surface ($\beta = 60°$, $\gamma = 25°$): full $\cos\theta$ equation (below) gives $\theta = 23.3°$	The west-facing surface looks almost straight at the afternoon sun.
Sunrise on the surface: solve $\cos\theta(\omega) = 0 \Rightarrow \omega = -69.5° \Rightarrow$ **7:22 AM**; sunset on the surface: $\cos\theta > 0$ until the sun sets ⇒ **5:33 PM**	Before 7:22 the sun is behind the west-facing surface; in the evening the surface sees the sun until it sets.

Numerical 4: angle of incidence and geometric factor at Nirala (2023 1b)

Note: the printed equation is the special form for a *vertical* surface ($\beta = 90°$). For a surface tilted 25° the full Duffie-Beckman equation must be used:

$$\cos\theta = \sin\delta\sin\varphi\cos\beta - \sin\delta\cos\varphi\sin\beta\cos\gamma + \cos\delta\cos\varphi\cos\beta\cos\omega + \cos\delta\sin\varphi\sin\beta\cos\gamma\cos\omega + \cos\delta\sin\beta\sin\gamma\sin\omega$$

Step (write this)	Why / what it means
$n = 31 + 15 = 46$; $\delta = 23.45 \sin[360(284 + 46)/365] = -13.29°$; 11:00 ⇒ $\omega = -15°$; $\varphi = 22.8°$, $\beta = 25°$, $\gamma = +8°$	Input angles.
$\cos\theta = (-0.2299)(0.3875)(0.9063) - (-0.2299)(0.9219)(0.4226)(0.9903) + (0.9732)(0.9219)(0.9063)(0.9659) + (0.9732)(0.3875)(0.4226)(0.9903)(0.9659) + (0.9732)(0.4226)(0.1392)(-0.2588)$	Term by term.
$\cos\theta = -0.0807 + 0.0887 + 0.7854 + 0.1524 - 0.0148 = 0.931 \Rightarrow$ **$\theta = 21.4°$**	
$\cos\theta_z = \cos 22.8 \cos(-13.29) \cos(-15) + \sin 22.8 \sin(-13.29) = 0.7776 \Rightarrow \theta_z = 39.0°$	Horizontal surface.

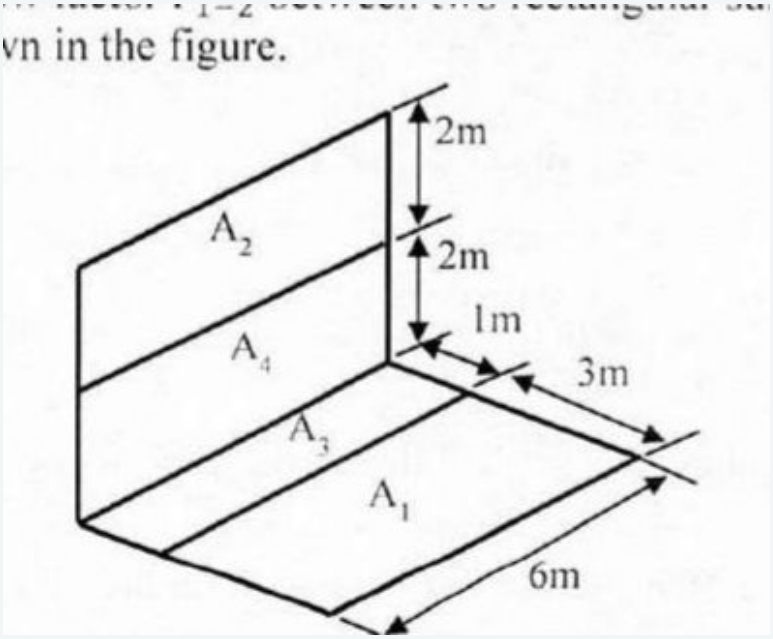
Geometric factor $R_b = \cos\theta/\cos\theta_z = 0.931/0.778 = 1.20$	The tilted surface receives 20% more beam radiation than a horizontal one.

Memory trick: "**Long-Day-Hour**": φ Latitude, δ Day (season), ω Hour. $\cos\omega_s = -\tan\varphi\tan\delta$; $N = 2\omega_s/15$. Solar time = clock + $4(L_{loc} - 90)$ + E for Bangladesh.

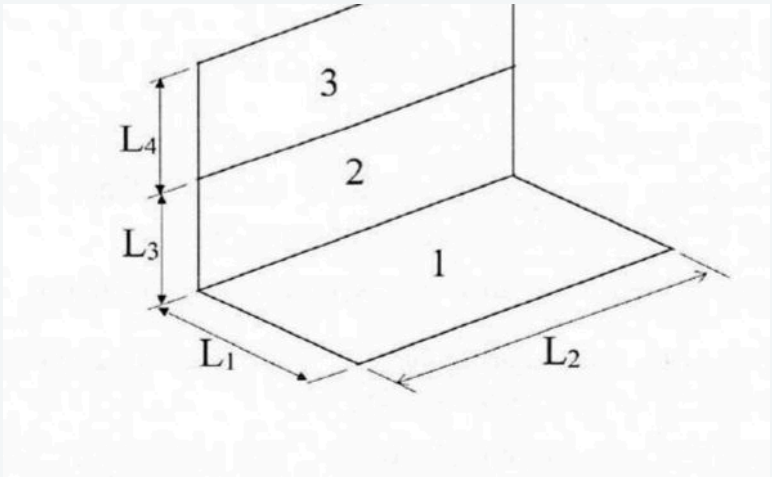
A4 View factors and radiation exchange: relations, networks, reradiating surface, shields Next to add Very likely · 100% Hard

PAST-PAPER QUESTIONS

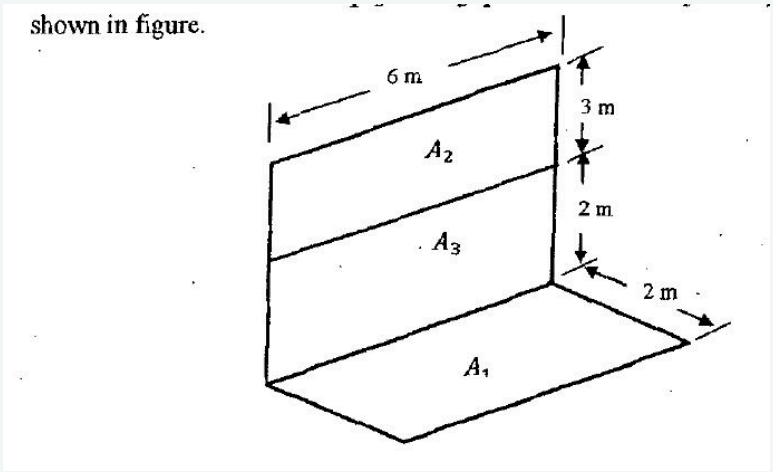
- **2023 Q3(a)** [5] Define view factor. Explain the crossed-strings method.
- **2023 Q3(b)** [10] Show that N radiation shields reduce heat transfer to 1/(N + 1) (all emissivities equal).
- **2023 Q3(c)** [12] Equilateral triangular duct, sides 1 m: base ε = 0.7 at 600 K, left side black at 1000 K, right side insulated. Heat supplied to the base per metre.
- **2022 Q3(d)** [10] View factor $F_{1\text{-}2}$ between the two rectangles in the figure.



- **2022 Q3(a)** [10] Net radiation exchange between two infinite parallel plates by the network method.
- **2022 Q3(b)** [9] What is a reradiating surface? Network for a three-zone enclosure with a reradiating surface.
- **2021 Q3(a)** [7] What is view factor? State the reciprocity relation.
- **2021 Q3(b)** [10] Net radiation exchange between two infinite parallel plates by the network method.
- **2021 Q3(c)** [5] What is a radiation shield? Why is it used?
- **2021 Q3(d)** [13] Parallel plates $\varepsilon_1 = 0.7$, $\varepsilon_2 = 0.5$, 900 K and 500 K: net radiation exchange.
- **2020 Q2(c)** [10] What is view factor? Deduce the expression for $F_{1\text{-}3}$ in the figure.



- **2020 Q3(a)** [10] What is radiosity? Radiation network for a three-zone enclosure with a reradiating surface.
- **2019 Q3(a)** [9] View factor; reciprocity and summation relations.
- **2019 Q3(d)** [9] View factors $F_{1\text{-}2}$ and $F_{2\text{-}1}$ for the perpendicular rectangles in the figure.



- **2019 Q3(c)** [8] Net heat flow in a two-zone enclosure.

LEARN IT FROM ZERO

Radiation networks, from zero

Radiation between real surfaces is solved just like an electric circuit. Each surface has a "battery" $E_b = \sigma T^4$. What actually leaves the surface is its **radiosity J** (emitted + reflected).

- **Surface resistance** $(1 - \varepsilon)/(A\varepsilon)$ between E_b and J: zero for a blackbody ($\varepsilon = 1$), so $J = E_b$.
- **Space resistance** $1/(A_1F_{12})$ between J_1 and J_2: how badly the surfaces see each other.

Two parallel plates: three resistances in series, giving $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$. A **shield** adds two more surface resistances and one more space resistance, so with N shields of the same ε the heat drops to $1/(N + 1)$. A **reradiating** (insulated) surface has $J = E_b$ and zero net heat: it just bounces energy on.

CALCULATOR **Parallel plates with or without radiation shields (2021 3(d), 2023 3(b))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Plate resistance sum: $1/\varepsilon_1 + 1/\varepsilon_2 - 1 = 1 / \mathbf{0.7} + 1 / \mathbf{0.5} - 1 = \mathbf{2.4286}$
2. Shields add: $N(2/\varepsilon_s - 1) = \mathbf{0} \times (2 / \mathbf{0.1} - 1) = \mathbf{0.0000}$
3. Net exchange: $q = \sigma(T_1^4 - T_2^4)/\Sigma R = 5.67\times10^{-8} \times (\text{pow}(\mathbf{900},4) - \text{pow}(\mathbf{500},4)) / (\mathbf{2.429} + \mathbf{0}) = \mathbf{13859}$ W/m²
4. Without shields: $5.67\times10^{-8} \times (\text{pow}(\mathbf{900},4) - \text{pow}(\mathbf{500},4)) / \mathbf{2.429} = \mathbf{13859}$ W/m²
5. Reduction: $100 \times (1 - \mathbf{13860} / \mathbf{13860}) = \mathbf{0.0}$ %

Set all three emissivities equal and N = 1: q falls to exactly 1/2; N = 3 gives 1/4. That is the $1/(N + 1)$ proof of 2023 3(b).

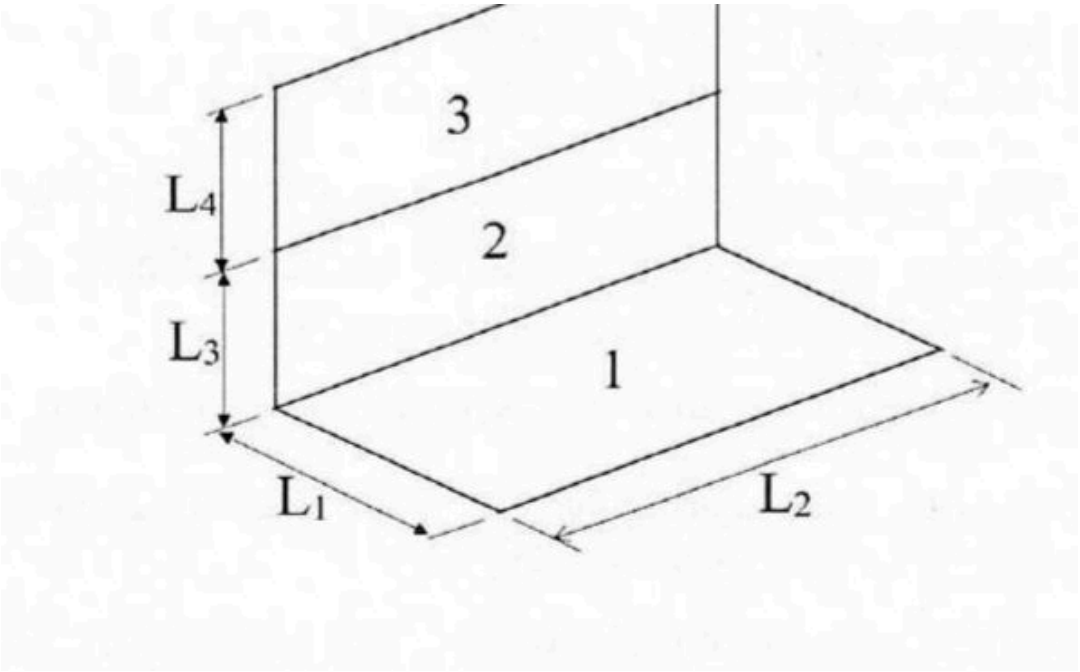
Before you start: view factors and networks

Radiation exchange between surfaces depends on their temperatures, emissivities and how well they "see" each other. The **view factor** F_{ij} is a purely geometric number: the fraction of radiation leaving surface i that strikes surface j directly. For gray diffuse surfaces the exchange is solved like an electric circuit with "potentials" $E_b = \sigma T^4$ and J.

View factor relations (2021 3a, 2019 3a, 2022 2d, 2023 3a)

1. **Reciprocity**: $A_i F_{ij} = A_j F_{ji}$ (follows from the double-integral definition, which is symmetric in i and j).
2. **Summation**: in an enclosure of N surfaces, $\Sigma_{j=1}^{N} F_{ij} = 1$ (all radiation leaving i lands somewhere); $F_{ii} = 0$ for a flat or convex surface.
3. **Superposition**: $F_{i\to(j,k)} = F_{ij} + F_{ik}$ (but not the reverse without reciprocity).
4. **Symmetry**: identical surfaces placed symmetrically with respect to i receive the same fraction.
5. **Crossed-strings method** (Hottel) for long 2-D geometries: imagine strings stretched between the edges; F_{12} = [Σ(crossed strings) − Σ(uncrossed strings)]/(2 × string on surface 1). It works for any shape, even with partial obstructions.

Expression for $F_{1\text{-}3}$ (2020 2c)

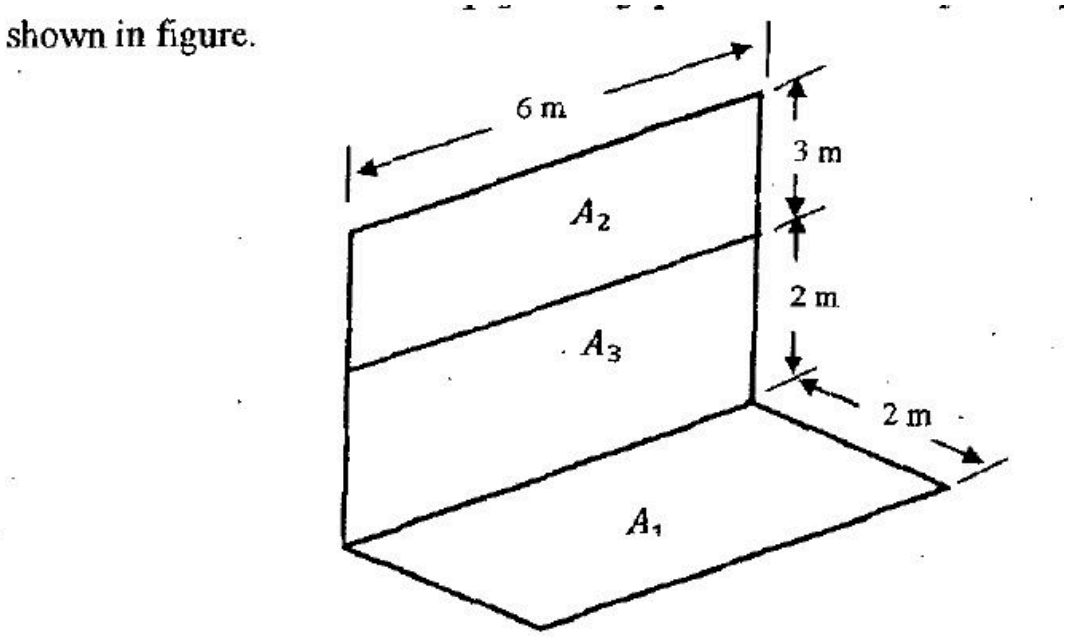


Question figure (2020 paper)

Step (write this)	Why / what it means
Surfaces 2 and 3 together form one rectangle (2,3) of height $L_3 + L_4$, perpendicular to 1 with common edge L_2	Use superposition.
$F_{1\to(2,3)} = F_{1\text{-}2} + F_{1\text{-}3}$	
$F_{1\text{-}3} = F_{1\to(2,3)} - F_{1\text{-}2}$	Both terms are perpendicular rectangles with a common edge, read from the chart (Cengel Fig. 13-6) with L_2 as the common edge: $F_{1\to(2,3)}$ at Y/X = L_1/L_2, Z/X = $(L_3 + L_4)/L_2$; $F_{1\text{-}2}$ at Z/X = L_3/L_2.

Reverse factor if needed: $F_{3-1} = A_1F_{1-3}/A_3 = (L_1/L_4)F_{1-3}$	Reciprocity.

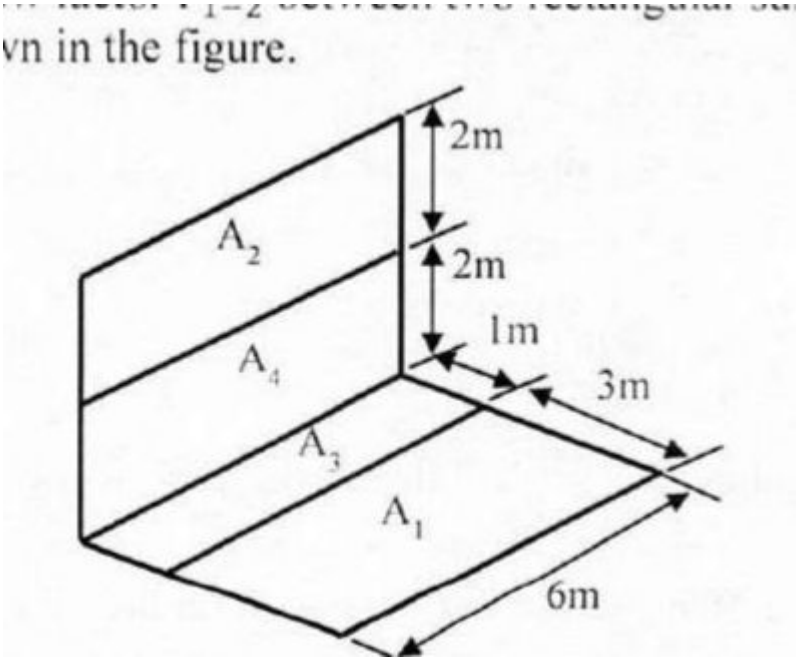
Numerical: F_{1-2} and F_{2-1} (2019 3d)



Question figure (2019 paper)

Step (write this)	Why / what it means
Common edge 6 m; $A_1 = 6 \times 2 = 12$ m^2 (horizontal); $A_3 = 6 \times 2$ m (vertical, next to the edge); $A_2 = 6 \times 3 = 18$ m^2 above A_3	Read the figure.
$F_{1\to(2,3)}$: $Y/X = 2/6 = 0.333$, $Z/X = 5/6 = 0.833 \Rightarrow 0.332$	Perpendicular-rectangle formula/chart (exact value).
F_{1-3}: $Y/X = 0.333$, $Z/X = 2/6 = 0.333 \Rightarrow 0.257$	
$F_{1-2} = 0.332 - 0.257 = 0.075$	Superposition.
$F_{2-1} = A_1F_{1-2}/A_2 = 12 \times 0.075/18 = 0.050$	Reciprocity. Chart readings give the same within ±0.01.

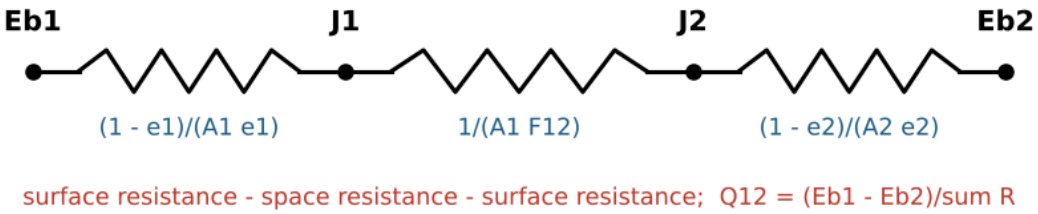
Numerical: $F_{1\text{-}2}$ with gaps on both surfaces (2022 3d)



Question figure (2022 paper)

Step (write this)	Why / what it means
Common edge 6 m. Horizontal: $A_3 = 6 \times 1$ (at the edge), $A_1 = 6 \times 3 = 18\ m^2$. Vertical: $A_4 = 6 \times 2$ (at the edge), $A_2 = 6 \times 2$ above it	Read the figure.
$A_1F_{12} = A_{13}F_{13\to24} - A_3F_{3\to24} - A_{13}F_{13\to4} + A_3F_{3\to4}$	Apply superposition twice (1+3 and 2+4) and reciprocity; every term is a common-edge pair.
Chart/formula values (X = 6): $F_{13\to24}$ (Y = 4, Z = 4) = 0.226; $F_{3\to24}$ (Y = 1, Z = 4) = 0.390; $F_{13\to4}$ (Y = 4, Z = 2) = 0.159; $F_{3\to4}$ (Y = 1, Z = 2) = 0.349	
$A_1F_{12} = 24(0.226) - 6(0.390) - 24(0.159) + 6(0.349) = 5.416 - 2.338 - 3.828 + 2.096 = 1.345\ m^2$	
$F_{1\text{-}2} = 1.345/18 = 0.075$	

Network method: two infinite parallel plates / two-zone enclosure (2021 3b, 2022 3a, 2019 3c)



Radiation network for two surfaces

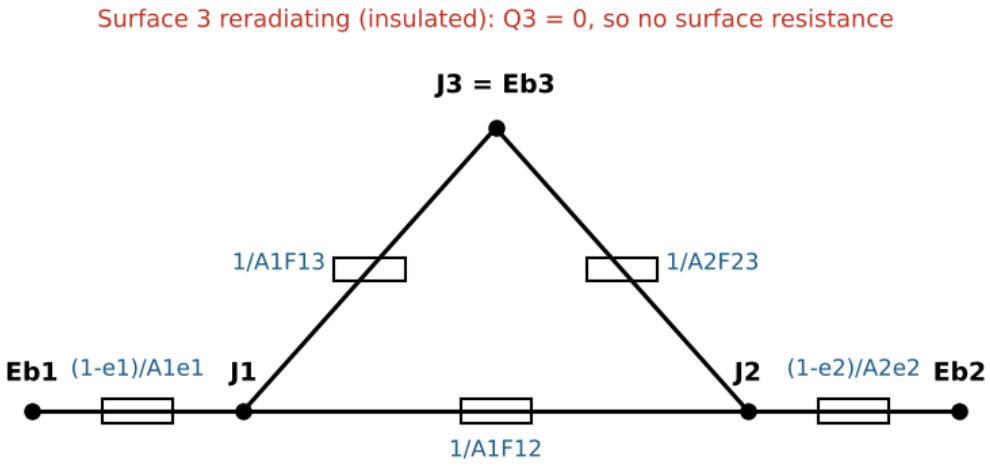
Step (write this)	Why / what it means

Net radiation leaving a gray diffuse surface: $Q_i = A_i(J_i - G_i)$ with $J_i = \varepsilon_i E_{bi} + (1 - \varepsilon_i)G_i$	Opaque: $\rho = 1 - \varepsilon$.
Eliminate G: $\mathbf{Q_i = (E_{bi} - J_i)/[(1 - \varepsilon_i)/(A_i\varepsilon_i)]}$	Surface resistance.
Exchange between i and j: $\mathbf{Q_{ij} = A_iF_{ij}(J_i - J_j) = (J_i - J_j)/(1/A_iF_{ij})}$	Space resistance (radiation leaving i that reaches j minus the reverse, with reciprocity).
Two-zone enclosure: three resistances in series $\Rightarrow Q_{12} = (E_{b1} - E_{b2})/[(1 - \varepsilon_1)/(A_1\varepsilon_1) + 1/(A_1F_{12}) + (1 - \varepsilon_2)/(A_2\varepsilon_2)]$	General two-surface result.
Infinite parallel plates: $A_1 = A_2 = A$, $F_{12} = 1$	
$\mathbf{q_{12} = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)}$	Divide by A: $(1 - \varepsilon_1)/\varepsilon_1 + 1 + (1 - \varepsilon_2)/\varepsilon_2 = 1/\varepsilon_1 + 1/\varepsilon_2 - 1$.

Numerical: parallel plates (2021 3d)

Step (write this)	Why / what it means
$q = 5.67\times10^{-8}(900^4 - 500^4)/(1/0.7 + 1/0.5 - 1)$	Formula above.
$= 5.67\times10^{-8} \times (6.561\times10^{11} - 0.625\times10^{11})/2.4286 = 33{,}658/2.4286$	
$\mathbf{q_{12} = 13{,}860\ W/m^2 \approx 13.9\ kW/m^2}$ from plate 1 to plate 2	

Three-zone enclosure with a reradiating surface (2020 3a, 2022 3b)



Network of a three-surface enclosure; surface 3 reradiating

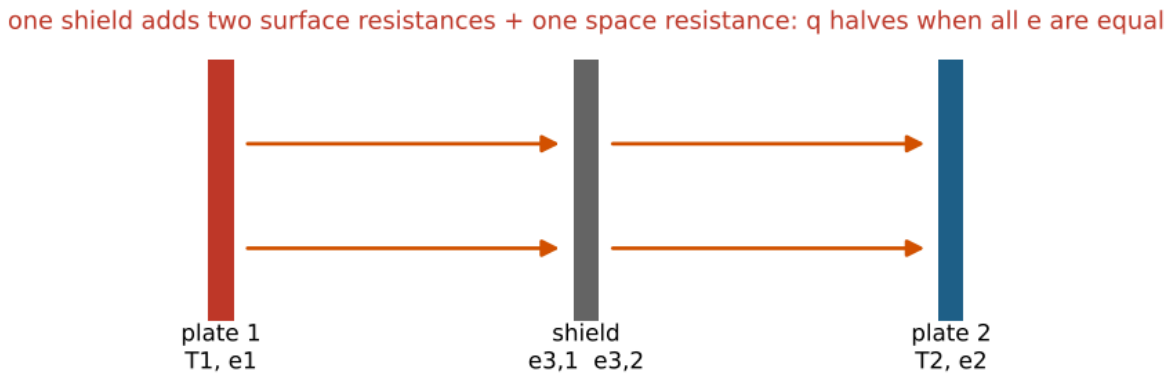
A **reradiating surface** is well insulated (adiabatic) and has zero net radiation ($Q_3 = 0$); it absorbs and re-emits everything, so $J_3 = E_{b3}$ and its surface resistance carries no current. Its temperature is set by the other surfaces. Network: E_{b1} — $(1 - \varepsilon_1)/A_1\varepsilon_1$ — J_1; J_1–J_2 through $1/A_1F_{12}$ *in parallel with* the path J_1–J_3–J_2 $(1/A_1F_{13} + 1/A_2F_{23})$; then J_2 — $(1 - \varepsilon_2)/A_2\varepsilon_2$ — E_{b2}:

$$Q_{12} = (E_{b1} - E_{b2})/\{\{R_1 + [A_1F_{12} + 1/(1/A_1F_{13} + 1/A_2F_{23})]^{-1} + R_2\}\}$$

Numerical: triangular duct (2023 3c)

Step (write this)	Why / what it means
Per metre length $A_1 = A_2 = A_3 = 1\ m^2$; equilateral triangle ⇒ $F_{ij} = 0.5$ for every pair	Symmetry + summation ($F_{ii} = 0$).
1 = base (ε = 0.7, 600 K), 2 = left (black, 1000 K: $J_2 = E_{b2}$), 3 = right (insulated: reradiating)	
$E_{b1} = \sigma 600^4 = 7348\ W/m^2$; $E_{b2} = \sigma 1000^4 = 56{,}700\ W/m^2$; $R_1 = 0.3/0.7 = 0.4286$	
Space network: $1/A_1F_{12} = 2$ in parallel with (2 + 2) = 4 ⇒ $R_{eq} = 1/(1/2 + 1/4) = 1.333$	
$Q_1 = (E_{b1} - E_{b2})/(R_1 + R_{eq}) = (7348 - 56{,}700)/1.762 = -28{,}010\ W$	
$Q_1 = -28.0$ kW per metre	Negative: the base *receives* 28 kW/m net radiation, so to keep it at 600 K this heat must be **removed** from the base (the question's "supplied" means supplied to the base by radiation). Right wall settles at $J_3 = 38{,}026\ W/m^2$ ⇒ $T_3 = 905$ K.

Radiation shields (2021 3c, 2023 3b)



One radiation shield between two plates

A **radiation shield** is a thin, highly reflective (low ε) sheet placed between two surfaces to reduce radiation heat transfer: it adds surface and space resistances. Uses: cryogenic vessels and multilayer insulation in spacecraft, thermocouple shields (to read true gas temperature), fire protection, furnace walls.

Step (write this)	Why / what it means
N shields between plates 1 and 2: N + 1 gaps; all ε = 1 so each gap has only a space resistance 1/A (F = 1)	Black shields: surface resistances vanish.
Steady: same q through every gap: $q = \sigma(T_1^4 - T_{s1}^4) = \sigma(T_{s1}^4 - T_{s2}^4) = \ldots = \sigma(T_{sN}^4 - T_2^4)$	Shields neither generate nor store energy.
Add the N + 1 equations: $(N + 1)q = \sigma(T_1^4 - T_2^4)$	Intermediate temperatures cancel.

$q_{N\ shields} = \sigma(T_1^4 - T_2^4)/(N + 1) = q_{no\ shield}/(N + 1)$	The same result holds whenever all emissivities are equal (each gap has the same total resistance).

Memory trick: "**Reciprocity, Summation, Superposition**" (RSS). Surface R = (1 − ε)/Aε; space R = 1/AF. Plates: $1/\varepsilon_1 + 1/\varepsilon_2 - 1$. Reradiating: $J = E_b$. N shields: ÷(N + 1).

Section B: Collector performance and applications

B4 Other applications: drying, desalination (solar still), cooling, pool vs pond, solar thermal power

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q7(c)** [10] Advantages of solar crop drying; working principle with schematic.
- **2023 Q8(d)** [9] Basic construction of a PV (photovoltaic) crop-drying system.
- **2023 Q8(a)** [11] Explain any one solar thermal process for cooling.
- **2023 Q8(b)** [6] Differences between solar pool heating and solar pond heating.
- **2023 Q8(c)** [9] Central receiver system to generate electricity.
- **2023 Q7(d)** [7] Briefly explain solar distillation.
- **2022 Q8(a)** [13] Solar crop drying; indirect crop-drying system with schematic.
- **2022 Q8(b)** [10] Solar-assisted liquid desiccant cooling system.
- **2022 Q8(c)** [12] Solar pond and solar thermal power; solar power tower with schematic.
- **2021 Q7(c)** [5] How to overcome limitations of open sun drying.
- **2021 Q7(d)** [12] Solar-assisted desiccant cooling system.
- **2021 Q8(a)** [8] Differences between solar pool heating and solar pond.
- **2021 Q8(b)** [14] How collectors generate electricity; parabolic dish system.
- **2021 Q8(c)** [13] What is solar desalination? Still with water 50 °C, glass 40 °C, h_{cw} = 2.40: hourly distillate per m^2.
- **2020 Q6(a)** [12] Solar pool heating; direct solar drying system with sketch.
- **2020 Q6(b)** [8] Central receiver system for thermal power generation.
- **2020 Q6(c)** [10] Still with water 60 °C, glass 50 °C, C = 0.04, n = 0.35, h_{cw} = 2.8: evaporative heat transfer per m^2.
- **2019 Q8(a)** [10] What is a solar pond? Working principle.
- **2019 Q8(b)** [13] Collectors for solar thermal power; central receiver and dish systems.
- **2019 Q8(c)** [12] What is distillation? Solar distillation with diagram.

★ REMEMBER IT

- Solar still: sun heats shallow salty water → water evaporates → vapour condenses on the cooler sloped glass → fresh droplets run down to a channel. Salt stays behind. Output only 2–5 L/m² per day.
- Solar pond: salt gradient (salty at the bottom) stops convection, so the bottom layer stores heat at 70–90 °C. Pool heating: just warms a pool, no gradient, no storage.
- Solar power: **trough** (line focus, one-axis), **tower** (heliostats to a central receiver, two-axis), **dish** (point focus with Stirling engine). All feed a heat engine.

CALCULATOR **Solar still: evaporative heat and distillate (2021 8(c), 2020 6(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. $P_w = \exp(25.317 - 5144/T_w)$: exp(25.317 - 5144 / (**50** + 273)) = **11984** Pa

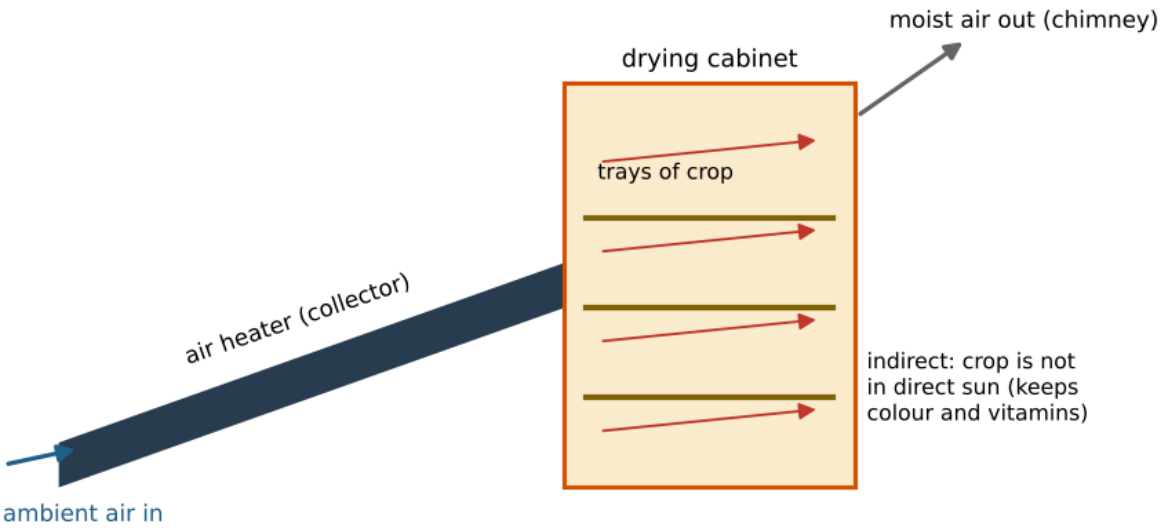
2. P_g: exp(25.317 - 5144 / (**40** + 273)) = **7205** Pa
3. Evaporative heat q_{ew}: $q_{ew} = 0.016273\ h_{cw}(P_w - P_g)$ = 0.016273 × **2.4** × (**11980** - **7205**) = **186.6** W/m²
4. Distillate per hour: $\dot{m} = q_{ew} \times 3600/\lambda$ = **186.6** × 3600 / (**2382** × 1000) = **0.282** kg/m²h

Use temperatures in kelvin inside the exponent. λ = 2382 kJ/kg at 50 °C, 2358 at 60 °C.

Before you start: using solar heat beyond hot water

Low-temperature solar heat (40-90 °C) dries crops, distils water, heats pools and drives desiccant/absorption coolers. High-temperature heat from concentrators (300-1000 °C) runs steam turbines or Stirling engines to make electricity. A salt-gradient solar pond is a collector and a store at once.

Solar crop drying (2022 8a, 2023 7c, 2021 7c, 2020 6a)



Indirect solar dryer

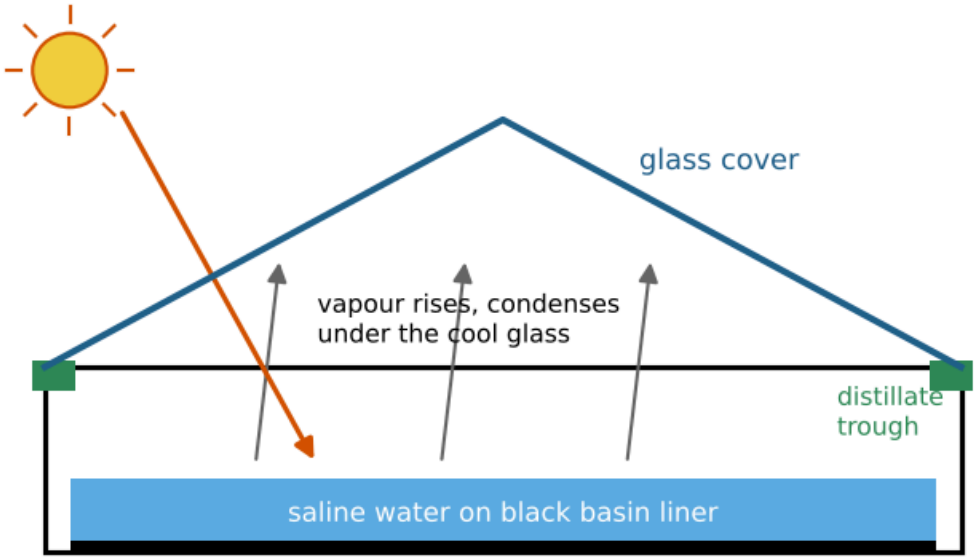
Solar crop drying removes moisture from agricultural products (grain, fruit, fish, spices) using solar-heated air to prevent spoilage and allow storage. **Problems of open sun drying:** contamination by dust, insects, birds and animals; re-wetting by rain and dew; overheating and loss of colour/vitamins by direct UV; slow, uneven drying and large space. **How solar dryers overcome them:** the crop is enclosed in a cabinet (clean, protected from rain and pests), drying air is heated in a collector (faster, controlled temperature), a chimney or fan gives airflow (uniform drying), and indirect dryers keep the crop out of direct sun (quality).

Direct (cabinet) dryer	Indirect dryer
Crop in a glazed box: absorbs sunlight directly; air enters at the bottom, leaves through a vent at the top	Separate solar air heater warms the air, which flows (natural draught or fan) through trays of crop in an opaque drying chamber, then out of a chimney
Simple and cheap; small quantities	Better quality (no direct sun), better temperature control, larger capacity

Indirect dryer working procedure: ambient air enters the collector, is heated 10-30 K by the black absorber, rises into the drying chamber, passes over and through the trays, picks up moisture from

the crop and leaves as warm humid air through the chimney; the chimney effect keeps the flow going. **Advantages of solar drying:** free energy, better hygiene and product quality, faster drying, less loss, higher market value. **PV crop dryer (2023 8d):** a solar air-heating dryer (collector + drying chamber + chimney) in which DC fans powered by a PV panel (with charge controller and small battery) force the air flow; the fan speed rises with sunshine, matching the heat supply, so the system works off-grid and dries faster and more uniformly than natural convection.

Solar distillation (desalination) (2019 8c, 2023 7d, 2021 8c)



Single-basin solar still

Distillation is separating pure water from dissolved salts and impurities by evaporating it and condensing the vapour. **Solar desalination** uses solar energy to produce fresh water from saline or brackish water. **Solar still working principle:** a shallow basin with a black liner holds 2-10 cm of saline water under a sloping glass cover. Sunlight passes the glass and is absorbed by the water and liner; the water warms and evaporates; the vapour rises by natural convection and condenses on the cooler underside of the glass, releasing latent heat; the condensate runs down the slope into collection troughs and is drawn off as distillate. Salt stays in the basin and is flushed periodically. Output 2-5 L/m^2 per day.

$q_{ew} = 0.016273\ h_{cw}(P_w - P_g)$, $P = \exp[25.317 - 5144/(T + 273)]$ (Pa), $\dot{m}_w = 3600\ q_{ew}/\lambda$ (kg/m^{2}h)

Numerical 1: hourly distillate (2021 8c)

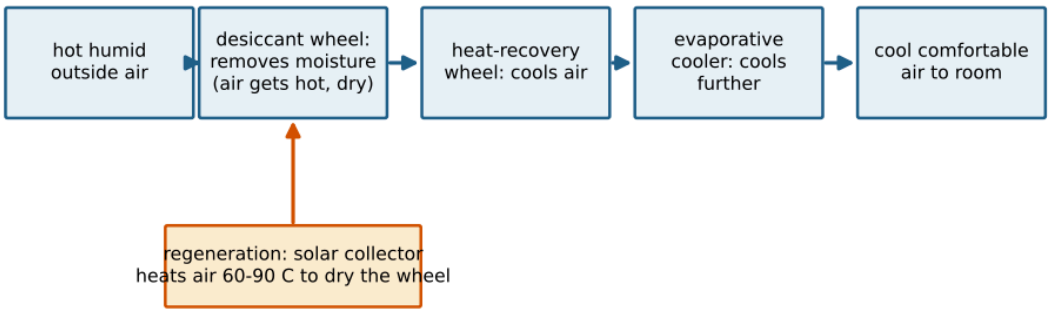
Step (write this)	Why / what it means
$P_w = \exp(25.317 - 5144/323) = 11{,}984$ Pa; $P_g = \exp(25.317 - 5144/313) = 7{,}205$ Pa	Partial vapour pressures at water (50 °C) and glass (40 °C).
$q_{ew} = 0.016273 \times 2.40 \times (11{,}984 - 7{,}205) = 186.6$ W/m^2	Dunkle's evaporative heat transfer.
λ at 50 °C = 2382 kJ/kg (steam table)	
$\dot{m}_w = 186.6 \times 3600/2.382\times10^6 = 0.282$ kg/m^{2}h	

Numerical 2: evaporative heat transfer (2020 6c)

Step (write this)	Why / what it means
$P_w(60\ °C) = \exp(25.317 - 5144/333) = 19{,}333$ Pa; $P_g(50\ °C) = 11{,}984$ Pa	
$q_{ew} = 0.016273 \times 2.8 \times (19{,}333 - 11{,}984) = 335$ W/m^2	Heat carried by the evaporating water per m^2 of still.
Distillate $= 335 \times 3600/2.358\times10^6 = 0.51$ kg/m^{2}h	λ at 60 °C = 2358 kJ/kg.

Note: C = 0.04 and n = 0.35 are the constants of Nu = C(Gr·Pr)^n, used to *calculate* h_{cw}; since h_{cw} = 2.8 W/m^{2}K is given, they are not needed (they would be used if h_{cw} were not given).

Solar cooling (2021 7d, 2022 8b, 2023 8a)



Solar-assisted solid desiccant cooling (open cycle)

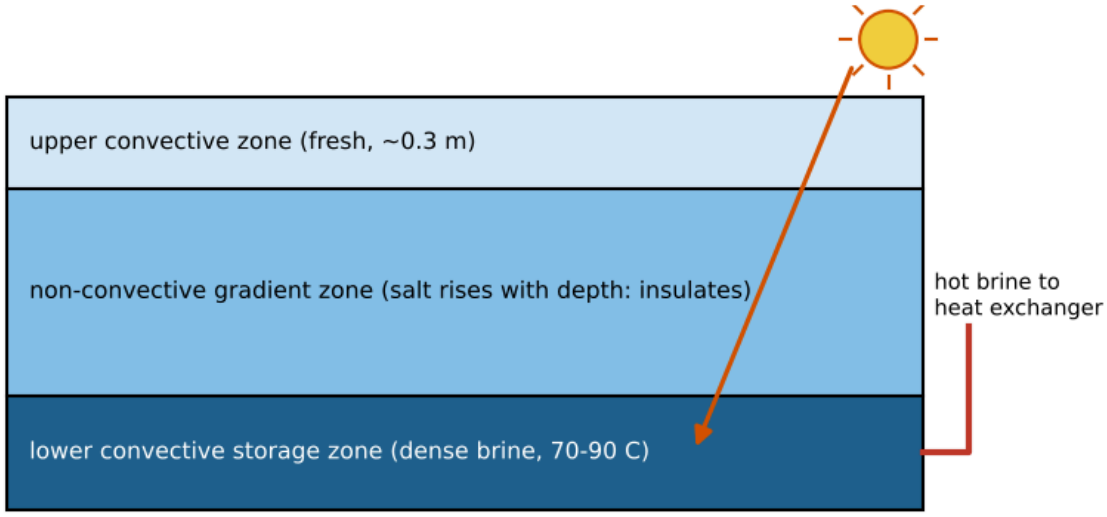
Solid desiccant cooling (2021 7d): hot humid outside air passes through a slowly rotating **desiccant wheel** (silica gel or zeolite) which adsorbs its moisture (the air becomes dry and warm); a **heat-recovery wheel** cools it using the cooler return air; an **evaporative cooler** then adds a little water, cooling it further to a comfortable supply temperature. The wheel is **regenerated** (dried) by return air heated to 60-90 °C by solar air or water collectors. No CFC refrigerant; runs on low-grade heat; handles humidity separately.

Liquid desiccant cooling (2022 8b): a concentrated salt solution (LiCl, $CaCl_2$, LiBr) is sprayed in an **absorber/dehumidifier** where it absorbs moisture from the supply air; the dry air is then cooled evaporatively. The diluted solution is pumped to a **regenerator** where solar-heated water or air (60-80 °C) evaporates the water and re-concentrates it; a solution heat exchanger saves heat; liquid desiccant can be stored (concentrated solution = energy storage).

Absorption cooling (2023 8a, any one process): LiBr-water machine: solar collectors (ETC, 80-95 °C) heat the **generator**, boiling water vapour out of the LiBr solution; vapour condenses in the **condenser**, passes an expansion valve and evaporates at low pressure in the **evaporator**, taking heat from chilled water (the cooling effect); the vapour is absorbed back into strong solution in the **absorber**, and a small pump returns the solution to the generator. The compressor is replaced by heat-driven absorption; COP 0.6-0.8.

Solar pool heating vs solar pond (2021 8a, 2023 8b, 2020 6a, 2019 8a)

Solar pool heating	Solar pond
Heats a swimming pool by circulating pool water (pool pump) through cheap unglazed plastic collectors	A large salt-water pond that is itself a collector and a long-term heat store
Low temperature rise (pool at 25-30 °C)	Bottom brine reaches 70-90 °C
Collectors separate from the water body; pool cover reduces night losses	No separate collector; salt gradient suppresses convection
Purpose: comfort, extends swimming season	Purpose: process heat, desalination, power (organic Rankine cycle), space heating



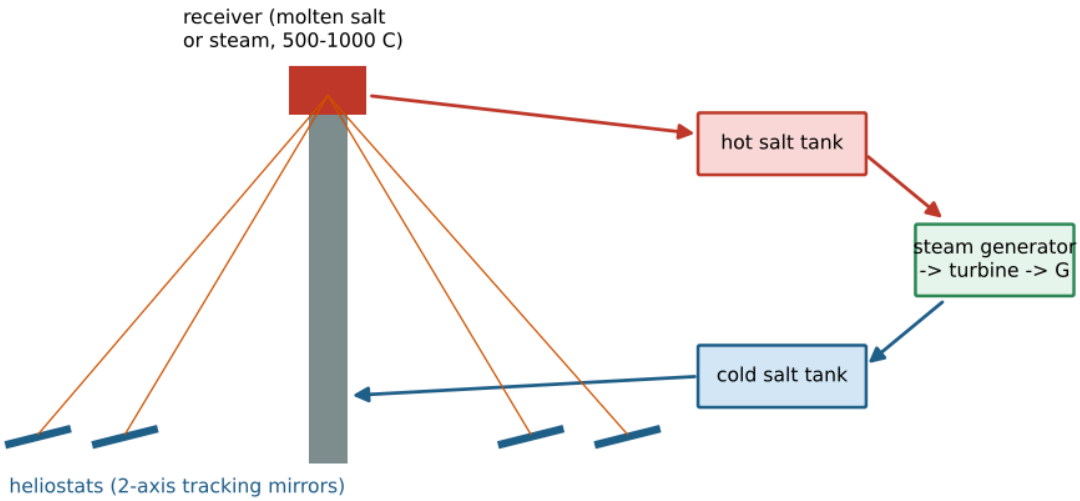
Salt-gradient solar pond

Solar pond working principle: three layers: a thin fresh **upper convective zone**; a **non-convective gradient zone** where salt concentration (and density) increases with depth; a dense **lower convective storage zone**. Sunlight penetrates to the bottom and heats the brine. In an ordinary pond hot water would rise and lose its heat at the surface, but here the hot bottom brine is so salty that it stays denser than the cooler water above, so **convection is suppressed**; the gradient zone acts as transparent insulation (heat can leave only by slow conduction). Heat is extracted by pumping hot brine through a heat exchanger. Area-intensive but cheap, with seasonal storage.

Solar pool heating and direct dryer (2020 6a)

Solar pool heating system: the pool filter pump draws water through the filter and then through unglazed black polypropylene collectors on the roof; a diverter valve (manual or automatic) sends water through the collectors only when they are warmer than the pool; warmed water returns to the pool; a pool blanket at night cuts evaporation loss. A **direct solar dryer** is a glazed, black-lined cabinet in which the crop on trays absorbs sunlight directly; air enters through holes at the bottom, is warmed, removes moisture and leaves through vents at the top.

Solar thermal power: central receiver, dish (2020 6b, 2022 8c, 2023 8c, 2019 8b, 2021 8b)



Central receiver (power tower) with molten-salt storage

Solar thermal power system: concentrating collectors convert sunlight into high-temperature heat that drives a heat engine (Rankine steam cycle, Brayton gas turbine or Stirling engine) to generate electricity. Collectors used: parabolic trough (most plants, 390 °C oil), linear Fresnel, central receiver with heliostats (565 °C molten salt), parabolic dish (Stirling, 750 °C).

Central receiver (power tower): a field of hundreds to thousands of **heliostats** (flat, two-axis tracking mirrors) reflects beam radiation onto a **receiver** at the top of a tall tower (C = 300-1500). The receiver heats a working fluid (molten nitrate salt, water/steam or air) to 500-1000 °C. Hot salt is stored in a hot tank; when power is needed it passes through a steam generator, and the steam drives a turbine-generator; the cooled salt returns to the cold tank. Storage lets the plant run after sunset. Example: Gemasolar (Spain), Crescent Dunes (USA).

Parabolic dish system: a paraboloid dish (or many mirror facets on a dish frame) tracks the sun on two axes and focuses beam radiation onto a receiver at its focal point (C = 500-3000), reaching 750 °C or more. A **Stirling engine** (or small gas turbine) mounted at the focus converts the heat directly into mechanical work, and an attached generator produces 5-25 kW. Highest solar-to-electric efficiency (~30%), modular, but costly and no simple storage.

> **Memory trick:** "**Dry, Distil, Desiccant, Dish**". Still: q_{ew} = 0.016273 $h_{cw}\Delta P$, ÷λ. Pond: "salt makes hot water heavy, so it stays down". Tower = heliostats + receiver + salt tanks + steam turbine.

B2 Concentrating collectors: definitions, classification, maximum concentration, tracking, materials

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q5(c)** [10] With figure, classify concentrating collectors.
- **2023 Q5(d)** [5] Advantages and disadvantages of concentrating collectors.
- **2023 Q6(c)** [13] Why is tracking essential? Types of tracking for concentrators.
- **2022 Q6(b)** [10] Define: aperture area, acceptance angle, concentration ratio, intercept factor.
- **2022 Q6(c)** [10] Concentration ratio vs receiver temperature; practical ranges.
- **2022 Q6(d)** [5] Characteristics a reflector should possess.
- **2021 Q6(b)** [14] With figures, classify solar concentrators.
- **2021 Q6(c)** [11] Why is tracking necessary? Advantages and disadvantages of concentrating collectors.
- **2020 Q5(a)** [10] What is solar tracking? Distinguish between "three-axis" and two-axis tracking.
- **2019 Q6(a)** [8] Define: aperture area, absorber area, acceptance angle, concentration ratio.
- **2019 Q6(d)** [11] Show that C_{max} for a circular concentrator is 46,747.
- **2019 Q6(b)** [8] Advantages and disadvantages of concentrating collectors.
- **2019 Q6(c)** [8] Why is tracking needed? Materials used in concentrators.

★ REMEMBER IT

- Concentration ratio C = aperture area / absorber area. Maximum possible (sun's half-angle θ_s = 0.27°): **3-D (circular) $C_{max} = 1/\sin^2\theta_s \approx 45{,}000–46{,}700$; 2-D (linear) $= 1/\sin\theta_s \approx 212$**.
- Higher C → smaller hot receiver → less loss → higher temperature: CPC (compound parabolic collector) 1–10 (< 150 °C), trough 15–45 (to 400 °C), dish 100–1000, tower 300–1500 (to 2000 °C).
- Concentrators use only **beam** radiation, so they must **track** the sun: one-axis for troughs, two-axis for dishes and heliostats.

CALCULATOR **Maximum concentration ratio (2019 6(d))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Circular (3-D): $C_{max} = 1/\sin^2\theta_s$ = 1 / pow(sin(**0.27**),2) = **45032**
2. Linear (2-D): $C_{max} = 1/\sin\theta_s$ = 1 / sin(**0.27**) = **212.2**

0.27° gives 45,036; the book figure 46,747 comes from θ_s = 0.265° (try it).

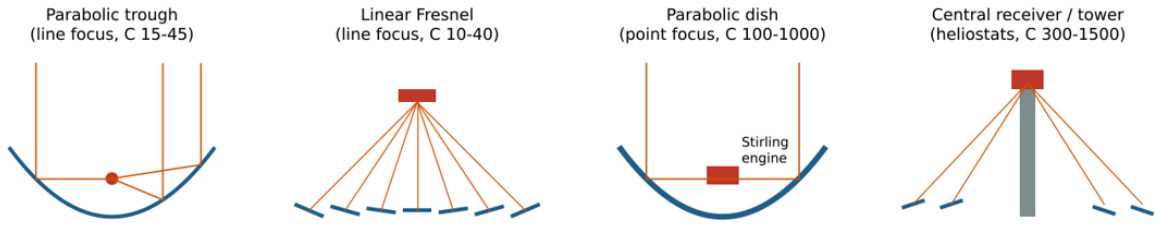
Before you start: why concentrate

Heat loss is proportional to the absorber area. A concentrator collects sunlight on a large aperture (mirror or lens) and focuses it onto a small receiver, so the loss area is small and the receiver gets much hotter. The price: only **beam** radiation can be focused, so the collector must **track** the sun.

Definitions (2019 6a, 2022 6b, 2021 6a, 2023 5a)

Term	Meaning
Aperture area A_a	the plane opening of the concentrator through which the solar radiation enters (the projected area of the reflector or lens facing the sun)
Absorber (receiver) area A_r	the area of the receiver that absorbs the concentrated radiation and delivers heat to the fluid
Concentration ratio C	geometric: $C = A_a/A_r$; optical: ratio of average flux on the receiver to the flux on the aperture
Acceptance angle $2\theta_c$	the angular range within which radiation entering the aperture reaches the receiver without moving the collector; θ_c = semi (half) acceptance angle. Large $2\theta_c$ ⇒ only seasonal adjustment; small ⇒ continuous tracking
Intercept factor γ	fraction of the reflected (focused) beam radiation that actually falls on the receiver; depends on receiver size, mirror errors and tracking errors; typically 0.9-0.97
Optical efficiency η_o	fraction of beam radiation on the aperture that is absorbed by the receiver: $\eta_o = \rho\tau\alpha\gamma[(1 - A_f\tan\theta)\cos\theta]$

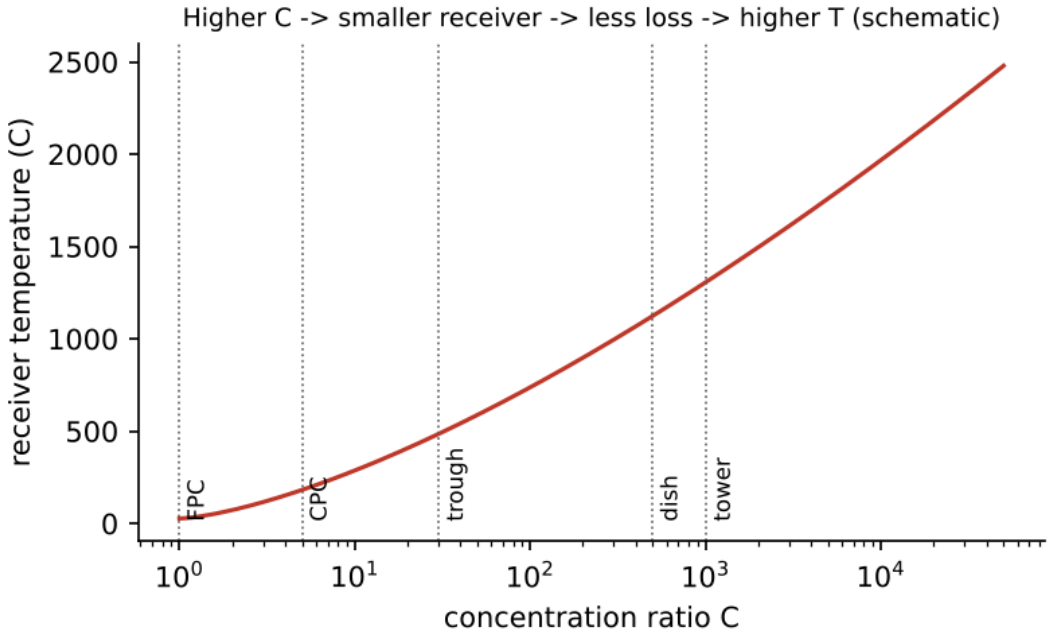
Classification of concentrators (2021 6b, 2023 5c)



Main concentrator types

1. **By focus:** *line focus* (2-D, cylindrical): parabolic trough collector (PTC), linear Fresnel reflector (LFR), cylindrical lens; *point focus* (3-D): parabolic dish, heliostat field + central receiver (tower), circular Fresnel lens.
2. **By optics:** reflecting (mirrors) or refracting (Fresnel lenses); imaging (forms an image of the sun: trough, dish) or non-imaging (CPC: collects all radiation within the acceptance angle, no image).
3. **By tracking:** stationary (CPC with large acceptance angle, flat plate with booster reflectors), single-axis tracking (trough, LFR), two-axis tracking (dish, heliostats).
4. **By receiver:** flat, cylindrical (tube, often in an evacuated glass envelope), cavity.

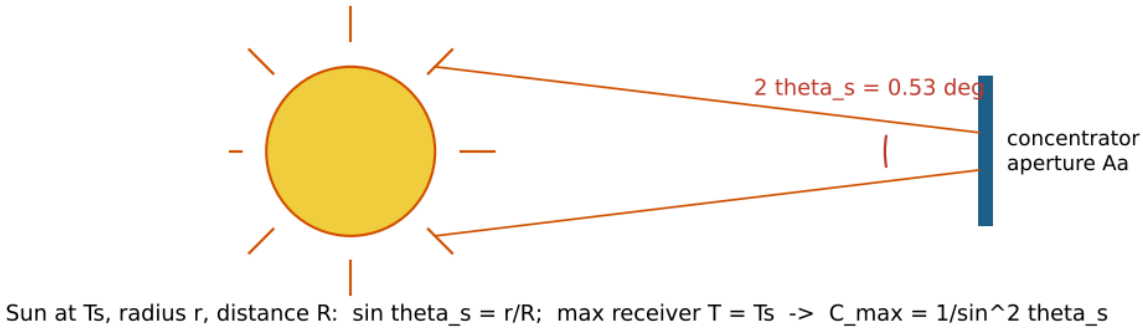
Concentration ratio and receiver temperature (2022 6c, 2023 5b)



Higher concentration gives higher receiver temperature

A receiver at temperature T_r absorbs $C\eta_o I_b$ per unit receiver area and loses $U_L(T_r - T_a) + \varepsilon\sigma T_r^4$; the higher C, the higher the temperature at which gain still exceeds losses (Rabl's diagram: the upper limit is the stagnation temperature, practical operation at 40-60% collection efficiency). Practical ranges: FPC (flat plate collector) C = 1 (< 80 °C); CPC 1-5 (to ~240 °C); linear Fresnel 10-40 and trough 15-45 (to 400 °C); dish 100-1000 (to 1500 °C); tower 100-1500 (to 2000 °C).

Maximum possible concentration ratio (2019 6d)



The sun seen from the concentrator subtends $2\theta_s = 0.53°$

Step (write this)	Why / what it means
Sun: radius r, temperature T_s, distance R; half-angle θ_s with $\sin\theta_s = r/R$	Geometry.
Radiation from the sun reaching the aperture: $Q_{s\to a} = A_a(r^2/R^2)\sigma T_s^4 = A_a\sin^2\theta_s\ \sigma T_s^4$	Sun emits $4\pi r^2\sigma T_s^4$, spread over a sphere of radius R.
Perfect (black) receiver radiates $A_r\sigma T_r^4$, of which a fraction $F_{r\text{-}s}$ reaches the sun: $Q_{r\to s} = A_r\sigma T_r^4 F_{r\text{-}s}$	No losses other than radiation back to the sun.

Second law: T_r cannot exceed T_s; at the limit $T_r = T_s$ and $Q_{s\to a} = Q_{r\to s} \Rightarrow A_a/A_r = F_{r\text{-}s}/\sin^2\theta_s$	
$F_{r\text{-}s} \le 1 \Rightarrow$ **$C_{max,3D} = 1/\sin^2\theta_s$** (circular), and for a linear concentrator **$C_{max,2D} = 1/\sin\theta_s$**	
Sun's half-angle $\theta_s \approx 16'$ (0.265°): $C_{max} = 1/\sin^2(0.265°) =$ **46,747** (circular); $1/\sin(0.265°) = 216$ (linear)	With θ_s rounded to 0.267° the value is 46,165; the course notes quote 46,747 (Kalogirou).

Tracking (2021 6c, 2019 6c, 2023 6c, 2020 5a)

Why tracking is necessary: a concentrator can focus only beam radiation that arrives within its small acceptance angle; as the sun moves (15°/h east-west and seasonally north-south), the focus would leave the receiver. Tracking keeps the aperture facing the sun so the image stays on the receiver; it also maximises the beam radiation collected ($\cos\theta \approx 1$).

Single-axis tracking	Two-axis tracking
Rotation about one axis; follows the sun in one direction only	Rotation about two perpendicular axes (azimuth-elevation or polar-declination); follows the sun exactly ($\theta = 0$ always)
Modes: horizontal E-W axis (adjust daily for altitude), horizontal N-S axis (tracks E-W during the day), polar axis parallel to the earth's axis (15°/h)	Used for point-focus systems: dishes, heliostats (each mirror), high-concentration PV (photovoltaic)
Simpler, cheaper, more reliable; for line-focus troughs and Fresnel	More complex and costly, higher maintenance; collects the most energy (~35-40% more than fixed)

The question's "three-axis" tracking is not a standard type; it is treated as single-axis vs two-axis. Tracking drives: mechanical/clock drives, microprocessor + sun-position algorithm (open loop), or light sensors with shadow bands (closed loop).

Advantages and disadvantages of concentrating collectors (2019 6b, 2023 5d, 2021 6c)

Advantages	Disadvantages
Higher fluid temperatures (to 400-1500 °C): power generation and process heat; better thermodynamic (Carnot) quality	Collect only beam radiation: useless on cloudy/hazy days, poor in humid climates
Small receiver area ⇒ smaller heat losses ⇒ higher efficiency at a given temperature	Need tracking systems: cost, power and maintenance
Reflectors are cheaper per m^2 than absorbers; less expensive receiver material	Reflecting surfaces lose reflectance with dust/ageing; need cleaning and refurbishment
Smaller heat-capacity mass ⇒ faster start-up; selective coatings/vacuum economical on small receivers	Non-uniform flux on the receiver; high temperatures cause material problems

Materials (2019 6c, 2023 4c): reflectors: back-silvered low-iron glass, polished or anodised aluminium (Alanod), silvered acrylic/polymer films; receivers: stainless or carbon steel tubes with cermet selective coatings in evacuated borosilicate glass envelopes; heat transfer fluids: synthetic oil, molten nitrate salt, water/steam; structures: galvanised steel. **A good reflector (2022 6d)** must have: high specular (mirror-like) reflectance over the solar spectrum (> 0.9); accurate shape;

resistance to weather, UV, abrasion and corrosion; low soiling and easy cleaning; light weight, rigidity and low cost; long life.

> **Memory trick:** "**$C = A_a/A_r$; $C_{max} = 1/\sin^2\theta_s \approx 46{,}747$**". Line focus (trough, Fresnel) = 1 axis; point focus (dish, tower) = 2 axes. Concentrators: "hot, small, but blind to diffuse".

B3 Solar water heating and storage: thermosiphon, ICS, direct/indirect, heat pump, storage tank numericals

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [10] Working principle of a direct solar water heating system.
- **2023 Q7(b)** [8] Problems of ICS (integrated collector storage) and possible solutions.
- **2022 Q7(b)** [8] Advantages and disadvantages of passive solar water heating.
- **2022 Q7(c)** [5] How to prevent reverse circulation in passive systems.
- **2022 Q7(a)** [12] Solar heat pump systems for water heating and space heating.
- **2022 Q7(d)** [10] 400 kg tank, UA = 10 W/°C, room 16 °C, 8 h from 7 AM, given Q_u and loads, initial 40 °C: final temperature.
- **2021 Q7(a)** [10] Working principle of an indirect solar water heating system.
- **2021 Q7(b)** [8] Integrated collector storage system; disadvantages.
- **2020 Q5(b)** [12] Classify solar water heating systems; thermosiphon system with sketch.
- **2020 Q5(c)** [8] How does a storage system work in a solar thermal system?
- **2019 Q7(a)** [7] What is solar water heating? Applications.
- **2019 Q7(b)** [10] Thermosiphon system with schematic.
- **2019 Q7(c)** [18] Why is storage necessary? 500 kg tank, UA = 12 W/°C, room 20 °C, 10 h from 5 AM, given Q_u and loads, initial 45 °C: final temperature.

★ REMEMBER IT

- **Hot rises, cold sinks: tank above collector** (thermosiphon, no pump). Direct = potable water goes through the collector; indirect = antifreeze loop plus heat exchanger (frost areas). ICS = tank is the collector (cheap, but big night losses).
- Tank energy balance each hour: **new T = old T + $[Q_u - L - UA(T - T_{room})\Delta t]/(mc_p)$**. Losses use the temperature at the *start* of the hour.

CALCULATOR **Storage tank: one hour at a time (2019 7(c), 2022 7(d))**

Change any number; every step is recalculated and shown with the numbers put in.

1. mc_p: **400** × 4180 / 10^6 = **1.6720** MJ/K
2. Tank loss this hour: UA × 3600(T − T_{room}) = **10** × 3600 × (**40** - **16**) / 10^6 = **0.864** MJ
3. Temperature at end of hour: T^+ = T + (Q_u − L − loss)/mc_p = **40** + (**0** - **10** - **0.864**) / **1.672** = **33.50** °C

Copy the end temperature into the start box, enter the next hour's Q_u and L, and repeat. This reproduces the hour-by-hour table above.

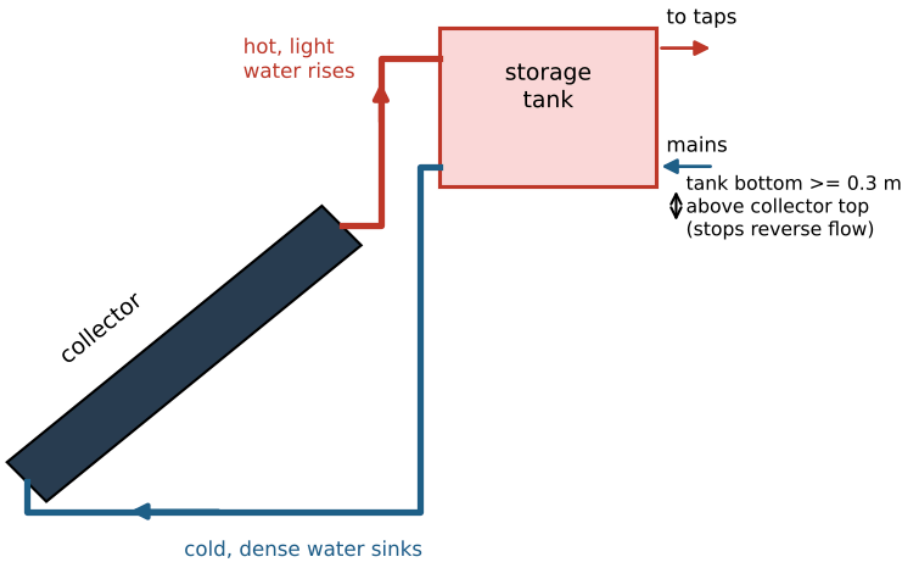
Before you start: solar water heaters

A solar water heater = collector + storage tank + a way to move water between them. If the water moves by itself (density difference), the system is **passive**; if a pump and controller move it, the system is **active**. Domestic hot water needs only 45-60 °C, so flat plate or evacuated tube collectors are used.

Classification (2020 5b, 2019 7a)

1. **Passive:** thermosiphon (natural circulation); integrated collector storage (ICS, batch heater).
2. **Active:** direct circulation (open loop), indirect circulation (closed loop, heat exchanger), drain-back, air-water systems, heat-pump systems.
3. **Applications (2019 7a):** domestic hot water for homes, hotels, hospitals, hostels; swimming-pool heating; space heating; industrial process heat (dairies, textiles, food); pre-heating boiler feed water; laundries.

Thermosiphon system (2020 5b, 2019 7b)



Thermosiphon solar water heater

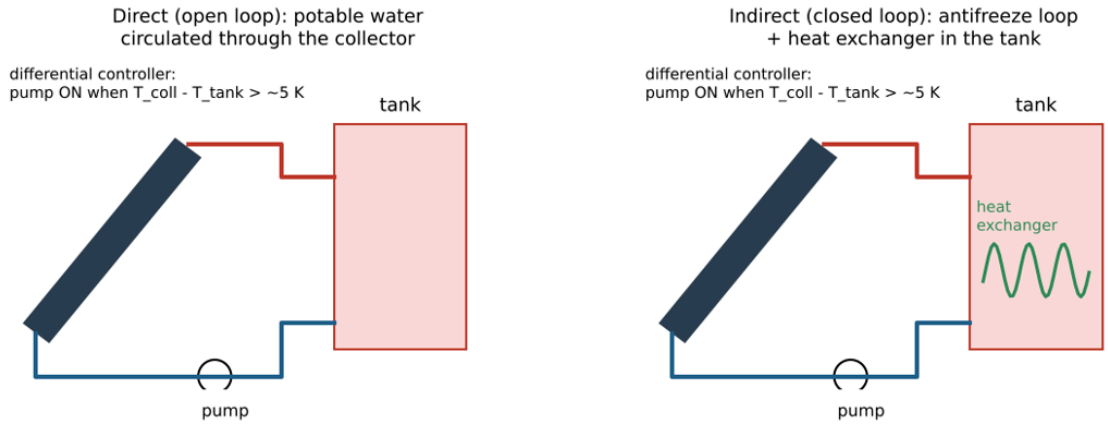
The storage tank is placed **above** the collector. Water in the collector absorbs heat, becomes lighter and rises through the upper pipe into the top of the tank; cooler, denser water from the bottom of the tank sinks through the lower pipe into the collector inlet. This natural (thermosiphon) circulation continues while the sun shines; the flow rate increases with the sunshine, so it is self-regulating. Hot water is drawn from the top of the tank and replaced by mains water at the bottom. An auxiliary electric heater near the top of the tank covers cloudy days. No pump or controller is needed.

Advantages and disadvantages of passive systems (2022 7b) and reverse circulation (2022 7c)

Advantages	Disadvantages
No pump, no controller, no electricity; works during power cuts	Tank must be above the collector: heavy load on the roof, poor appearance
Simple, cheap, reliable, long life, low maintenance	Low flow; performance depends on stratification and pipe layout; less flexible design
Self-regulating flow	Freezing risk in cold climates (open systems); scaling with hard water
	Reverse circulation at night cools the tank; ICS has high night losses

Preventing reverse circulation: at night the collector cools below the tank (sky radiation), and the flow can reverse, cooling the stored water. Remedies: place the bottom of the tank at least 0.3 m (often 0.3-0.6 m) above the top of the collector; fit a non-return (check) valve or thermosiphon restrictor in the circuit; use a heat-pipe collector (one-way heat flow); in active systems, the controller stops the pump and a check valve prevents back-flow.

Direct and indirect active systems (2021 7a, 2023 7a)



Direct (open-loop) and indirect (closed-loop) active systems

Direct (open loop): a pump circulates the potable water itself from the bottom of the tank through the collectors and back to the tank. A **differential thermostat** switches the pump on when the collector outlet is about 5-10 K hotter than the tank bottom and off when the difference drops to 1-2 K; a check valve stops night reverse flow. Simple and efficient (no heat-exchanger loss) but only suitable where there is no freezing and the water is soft.

Indirect (closed loop): a separate collector loop filled with antifreeze (water-propylene glycol) or another fluid is pumped through the collectors and through a **heat exchanger** (coil inside the tank or external). The heat passes to the potable water without mixing. An expansion tank and pressure-relief valve protect the closed loop. Used in freezing climates and with hard water; slightly lower efficiency due to the heat exchanger.

Integrated collector storage (2021 7b, 2023 7b)

In an **ICS** system the storage tank itself is the absorber: one or more black-painted (or selectively coated) tanks sit inside an insulated box with a glazed top; water is heated in place; hot water is

drawn from the top and cold water enters at the bottom. **Problems:** large tank surface loses heat to the surroundings, especially at night and in cold weather (the main drawback); water may freeze; little stratification; heavy. **Solutions:** selective coating on the tank, double glazing or transparent insulation, insulated night covers (manual or automatic), placing tanks in a CPC (compound parabolic collector)/involute reflector cavity, using the system for daytime loads, or adding a separate insulated storage tank.

Solar heat pump systems (2022 7a)

Water heating: a refrigerant circuit where the evaporator is the solar collector (direct-expansion, unglazed or glazed) or receives heat from solar-heated water; the compressor raises the temperature, and the condenser coil heats the storage tank. Solar input raises the evaporator temperature, so the COP (3-6) is higher than with an air-source heat pump. **Space heating:** collectors heat a storage tank; when the tank is warm enough (> ~25 °C) it heats the house directly; when it is too cool, a water-to-air heat pump uses the tank as its heat source (series system); a parallel system uses solar directly and an air-source heat pump as backup; dual-source systems choose the better source.

Storage: why and how (2019 7c, 2020 5c)

Solar input and demand do not coincide (sun at noon, hot-water demand morning and evening, none at night or on cloudy days), so energy must be **stored**. A storage system smooths supply, raises the solar fraction and lets the collector run at a lower temperature. **How it works:** during sunshine the collector loop delivers Q_u to the tank (sensible heat in water, rock bed for air systems, or latent heat in phase-change material); the load draws heat; the tank loses $UA(T_s - T_a)$ to its surroundings. Well-stratified tanks (hot at the top, cold at the bottom) return the coldest water to the collector and supply the hottest water to the load, improving performance.

$$T_s^+ = T_s + (\Delta t/mc_p)[Q_u - L - (UA)(T_s - T_a)] \quad \text{(fully mixed tank, explicit hour-by-hour)}$$

Numerical 1: 500 kg tank, 10 hours from 5 AM (2019 7c)

mc_p = 500 × 4180 = 2.09 MJ/K; tank loss per hour = 12 × 3600$(T_s - 20)$ J = 0.0432$(T_s - 20)$ MJ.

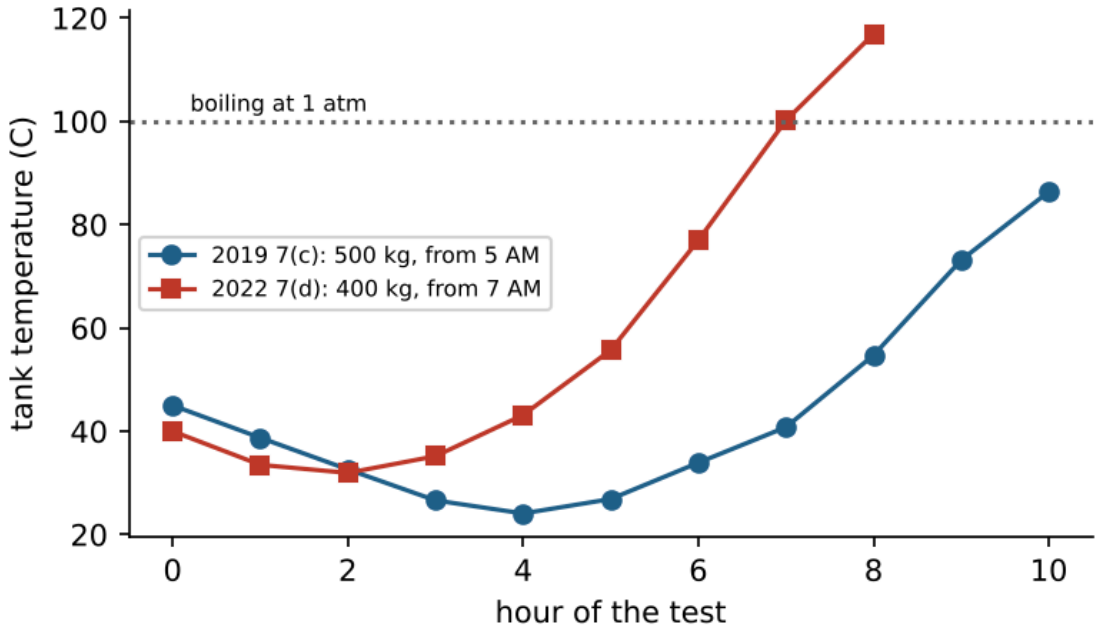
Hour	Q_u (MJ)	L (MJ)	Loss (MJ)	T_s at end (°C)
5-6	0	12	1.080	38.74
6-7	0	12	0.810	32.61
7-8	0	12	0.545	26.61
8-9	10	15	0.286	24.08
9-10	21	15	0.176	26.87
10-11	30	15	0.297	33.90
11-12	40	25	0.601	40.79
12-13	55	25	0.898	54.72
13-14	65	25	1.500	73.14
14-15	55	25	2.296	86.39

Step (write this)	Why / what it means
`First hour: T_s+ = 45 + [0 - 12 - 0.0432(45 - 20)]/2.09 = 45 - 13.08/2.09 = 38.74 °C`	Repeat with the new temperature each hour.
`Final tank temperature at 3 PM ≈ 86.4 °C`	

Numerical 2: 400 kg tank, 8 hours from 7 AM (2022 7d)

mc_p = 400 × 4180 = 1.672 MJ/K; loss per hour = 10 × 3600(T_s − 16) J = 0.036(T_s − 16) MJ.

Hour	Q_u (MJ)	L (MJ)	Loss (MJ)	T_s at end (°C)
7-8	0	10	0.864	33.50
8-9	8	10	0.630	31.93
9-10	20	14	0.573	35.17
10-11	28	14	0.690	43.14
11-12	40	18	0.977	55.71
12-13	55	18	1.430	76.98
13-14	65	24	2.195	100.19
14-15	55	24	3.031	116.92



Tank temperature hour by hour for both numericals

Result: final temperature = 116.9 °C at 3 PM. Flag: with the printed data the water passes 100 °C at 2 PM, which is possible only in a pressurised tank; an open tank would boil and the relief valve would dump heat. Write the hour-by-hour answer and add this remark.

> **Memory trick: "Hot rises, cold sinks: tank above collector"**. Tank step: **new T = old T + (gain − load − loss)/mc**, loss = UA(T − T_{room}) × 3600 s.

B1 Flat plate collector performance: losses, U_L, F', F_R, critical radiation level, testing, numericals

Next to add · Very likely · 100% · Hard

PAST-PAPER QUESTIONS

- **2023 Q5(a)** [8] Define: critical radiation level, collector efficiency factor, heat removal factor, semi-acceptance angle.
- **2023 Q5(b)** [12] With figures, significance of concentration ratio and effective (τα).
- **2023 Q6(a)** [12] Discuss all the losses of a flat plate collector.
- **2022 Q5(a)** [8] Significance of: critical radiation level, effective (τα), heat removal factor, inclination angle.
- **2022 Q5(b)** [15] Derive a general expression for the collector efficiency factor.
- **2022 Q6(a)** [10] Testing procedure for a new FPC (flat plate collector).
- **2022 Q5(c)** [12] 5 m^2 FPC tested in the dark: water 65 → 50 °C at 0.05 L/s, ambient 8 °C: U_L.
- **2021 Q6(a)** [10] Define: acceptance angle, concentration ratio, critical radiation level, collector efficiency factor, heat removal factor.
- **2021 Q5(a)** [10] Thermal network for a double-glass-covered FPC.
- **2021 Q5(c)** [19] U_L for an FPC at 45°: T_a = 20 °C, glass 50 °C, plate 70 °C, 5 cm glass wool (k = 0.03), air gap "75 cm", ε_g = 0.8, ε_p = 0.95.
- **2020 Q4(a)** [6] Define: concentration ratio, critical radiation level, collector efficiency factor.
- **2020 Q4(c)** [16] FPC 4 m^2, 0.06 kg/s, U_L = 6.9, W = 120 mm, D = 15/13.5 mm, plate 0.4 mm copper, h_{fi} = 320, τα = 0.8, 2.88 MJ/m^2 in 1 h, $T_i - T_a$ = 5 °C: F, F′, Q_u, η.
- **2019 Q4(b)** [16] Define and explain: critical radiation level, collector efficiency factor, effective (τα), heat removal factor.
- **2019 Q5(b)** [30] U_L for an 8 m × 2.5 m FPC at 45° (plate 69 °C, air 20 °C, gap 75 mm, ε_p = 0.95, ε_g = 0.88, wind 2.5 m/s, back 60 mm and edge 20 mm insulation k = 0.04, depth 0.08 m).

$$Nu = 1 + 1.44\left[1 - \frac{1708}{Ra\cos\beta}\right]^{+}\left(1 - \frac{1708[\sin(1.8)\beta]^{1.6}}{Ra\cos\beta}\right) + \left\{\left[\frac{Ra\cos\beta}{5830}\right]^{1/3} - 1\right\}^{+}$$

LEARN IT FROM ZERO

F′ and F_R, from zero

The ideal collector would be at the temperature of the incoming water everywhere, so it would lose the least heat. Reality has two penalties, each written as a factor below 1:

- **Fin efficiency F**: the plate between tubes is a fin, hotter in the middle than at the tube, F = tanh[m(W − D)/2]/[m(W − D)/2], m = √(U_L/kδ).
- **Collector efficiency factor F′**: the plate is hotter than the fluid next to it (fin + tube wall + inside film resistances). F′ = useful gain / gain if the plate were at the *local fluid* temperature.
- **Heat removal factor F_R**: the fluid also warms up as it flows. F_R = useful gain / gain if the whole plate were at the *inlet* temperature. F_R = F′F″, with flow factor F″ from $\dot{m}c_p$.

Then the Hottel-Whillier-Bliss equation: $\mathbf{Q_u = A_c F_R[S - U_L(T_i - T_a)]}$, $S = (\tau\alpha)I_T$. The **critical radiation level** is the I_T where $Q_u = 0$: $I_c = U_L(T_i - T_a)/(\tau\alpha)$.

CALCULATOR Flat plate collector: F, F′, F_R, Q_u, efficiency (2020 4(c))

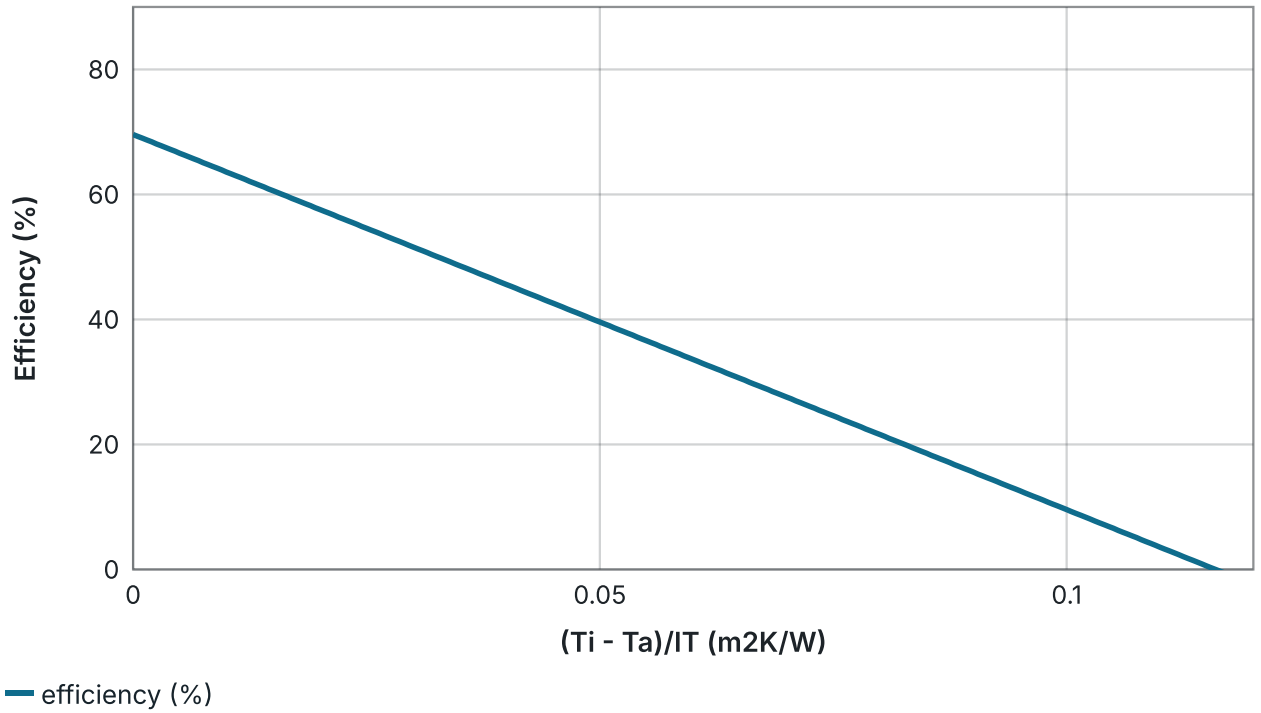
Change any number; every step is recalculated and shown with the numbers put in.

1. $m = \sqrt{(U_L/k\delta)}$: $\sqrt{(\mathbf{6.9} / (\mathbf{385} \times \mathbf{4.000 \times 10^{-4}}))} = \mathbf{6.694}$ 1/m
2. $m(W - D)/2$: $\mathbf{6.694} \times (\mathbf{0.12} - \mathbf{0.015}) / 2 = \mathbf{0.3514}$
3. Fin efficiency F: $F = \tanh x / x = \tanh(\mathbf{0.3514}) / \mathbf{0.3514} = \mathbf{0.961}$
4. F′: $F' = (1/U_L)/W[1/U_L(D + (W - D)F) + 1/\pi D_i h_{fi}] = (1 / \mathbf{6.9}) / (\mathbf{0.12} \times (1 / (\mathbf{6.9} \times (\mathbf{0.015} + (\mathbf{0.12} - \mathbf{0.015}) \times \mathbf{0.9608})) + 1 / (\pi \times \mathbf{0.0135} \times \mathbf{320}))) = \mathbf{0.912}$
5. Flow factor F″: $\mathbf{0.06} \times \mathbf{4180} / (\mathbf{4} \times \mathbf{6.9} \times \mathbf{0.9119}) \times (1 - \exp(-\mathbf{4} \times \mathbf{6.9} \times \mathbf{0.9119} / (\mathbf{0.06} \times \mathbf{4180}))) = \mathbf{0.951}$
6. $F_R = F'F''$: $\mathbf{0.9119} \times \mathbf{0.9515} = \mathbf{0.868}$
7. Absorbed $S = (\tau\alpha)I_T$: $\mathbf{0.8} \times \mathbf{800} = \mathbf{640.0}$ W/m²
8. Useful gain: $Q_u = A_c F_R[S - U_L(T_i - T_a)] = \mathbf{4} \times \mathbf{0.8677} \times (\mathbf{640} - \mathbf{6.9} \times \mathbf{5}) = \mathbf{2102}$ W
9. Efficiency: $100 \times \mathbf{2102} / (\mathbf{4} \times \mathbf{800}) = \mathbf{65.7}$ %
10. Critical radiation level: $\mathbf{6.9} \times \mathbf{5} / \mathbf{0.8} = \mathbf{43.1}$ W/m²

If the paper gives hourly irradiation in MJ/m², divide by 3600 s for W/m² (2.88 MJ/m²h = 800 W/m²).

INTERACTIVE Collector efficiency line (test result)

Move the sliders and watch the curve change.



Intercept $F_R(\tau\alpha)$: **69.6** % · Slope $-F_RU_L$: **-6.00** W/m²K · Zero efficiency at: **0.1159** m²K/W

Testing (2022 6(a)): measure η at several inlet temperatures, plot against $(T_i - T_a)/I_T$, fit a straight line. Better collector = higher intercept, flatter slope.

CALCULATOR **U_L from a night (dark) test (2022 5(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Mass flow: **0.05** × **0.98** = **0.0490** kg/s
2. Heat lost: $Q = \dot{m}c_p(T_i - T_o)$ = **0.049** × 4180 × (**65** - **50**) = **3072** W
3. Mean fluid temperature: (**65** + **50**) / 2 = **57.5** °C
4. U_L: $U_L = Q/[A_c(T_m - T_a)]$ = **3072** / (**5** × (**57.5** - **8**)) = **12.41** W/m²K

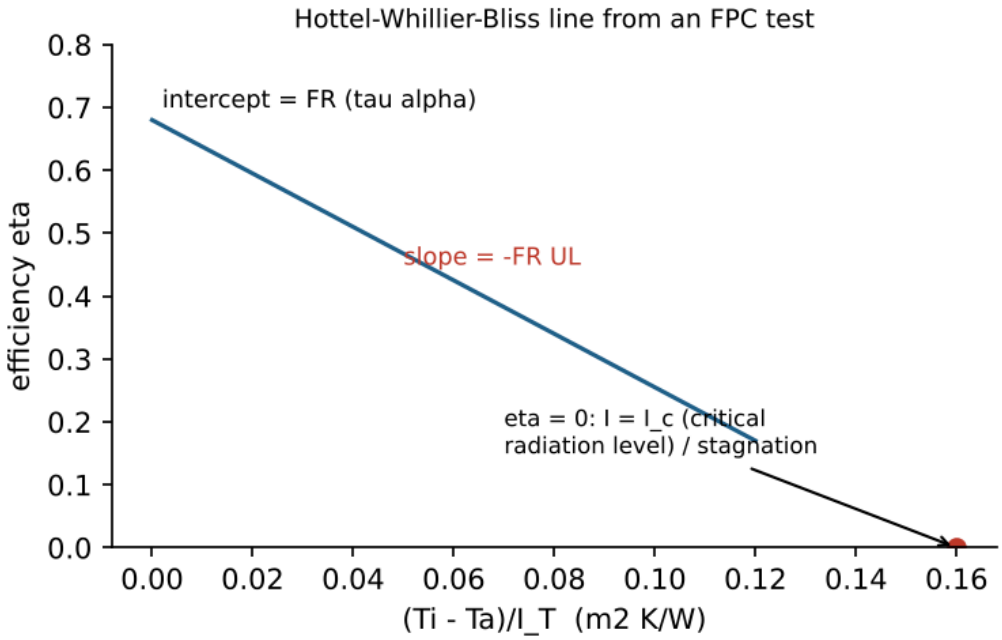
Before you start: the collector energy balance

Of the sunlight I_T falling on the collector, a fraction $(\tau\alpha)$ is absorbed: $S = (\tau\alpha)I_T$. Part of it is lost because the plate is hotter than the air: $U_L(T_p - T_a)$. The rest is the useful gain carried away by the fluid.

$Q_u = A_c[S - U_L(T_p - T_a)] = A_cF_R[S - U_L(T_i - T_a)] = \dot{m}c_p(T_o - T_i)$; $\eta = Q_u/(A_cI_T)$

Definitions and significance (2020 4a, 2021 6a, 2019 4b, 2022 5a, 2023 5a/5b)

Term	Definition	Significance
Critical radiation level I_c	the radiation level at which absorbed energy just equals losses, so useful gain is zero: $I_c = U_L(T_i - T_a)/(\tau\alpha)$	the collector produces useful heat only when $I_T > I_c$; the pump controller should not run below it; I_c rises with operating temperature
Collector efficiency factor F'	ratio of the actual useful energy gain to the gain if the whole absorber were at the *local fluid* temperature; $= (1/U_L)/(1/U_o)$, U_o = heat transfer coefficient from fluid to ambient	measures the quality of the plate-tube design (fin efficiency, bond, h_{fi}); typical 0.8-0.97; constant for a given design
Heat removal factor F_R	ratio of actual useful gain to the gain if the whole absorber were at the *fluid inlet* temperature: $F_R = (\dot{m}c_p/A_cU_L)[1 - \exp(-A_cU_LF'/\dot{m}c_p)] = F'F''$	lets Q_u be computed from the known inlet temperature; like a heat-exchanger effectiveness; rises with flow rate towards F'
Effective $(\tau\alpha)$	$(\tau\alpha) = \tau\alpha/[1 - (1 - \alpha)\rho_d]$ accounts for multiple reflections between plate and cover; $(\tau\alpha)_e \approx 1.01\text{-}1.02(\tau\alpha)$ also counts solar energy absorbed in the glass that comes back to the plate	sets the optical (zero-loss) efficiency $F_R(\tau\alpha)$, the intercept of the efficiency line; selective coatings and low-iron glass raise it
Inclination angle β	slope of the collector from the horizontal	maximises incident radiation: $\beta \approx$ latitude for year-round use, φ + 10-15° for winter, φ − 10-15° for summer; affects R_b, view of sky/ground and gap convection; thermosiphon/heat pipe need a minimum tilt
Concentration ratio C	A_a/A_r (see B2)	higher C ⇒ smaller lossy receiver ⇒ higher temperature



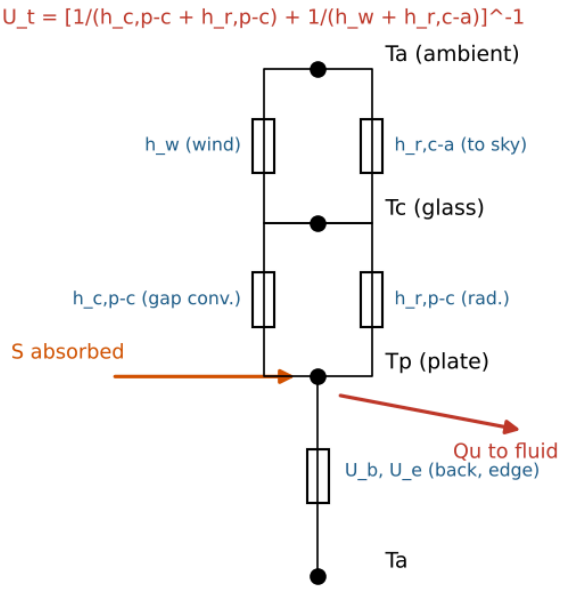
Collector efficiency line: intercept $F_R(\tau\alpha)$, slope $-F_RU_L$

Losses of a flat plate collector (2023 6a)

1. **Optical losses**: reflection at the glass surfaces (~8%), absorption in the glass, dust and shading by the frame, reflection from the absorber ($\alpha < 1$); grow at large incidence angles.
2. **Top loss** (largest thermal loss): natural convection and long-wave radiation from plate to glass, then wind convection and radiation from glass to sky; coefficient U_t.
3. **Bottom (back) loss**: conduction through the back insulation and convection/radiation from the casing: $U_b \approx k_i/L_i$.
4. **Edge loss**: conduction through the side insulation: $U_e = (k/L)_{edge} A_{edge}/A_c$.
5. **Other**: heat stored in the collector mass at start-up (heat capacity), pipe and connection losses, night-time reverse circulation.

$$U_L = U_t + U_b + U_e; \quad U_t = [1/(h_{c,p\text{-}c} + h_{r,p\text{-}c}) + 1/(h_w + h_{r,c\text{-}a})]^{-1}$$

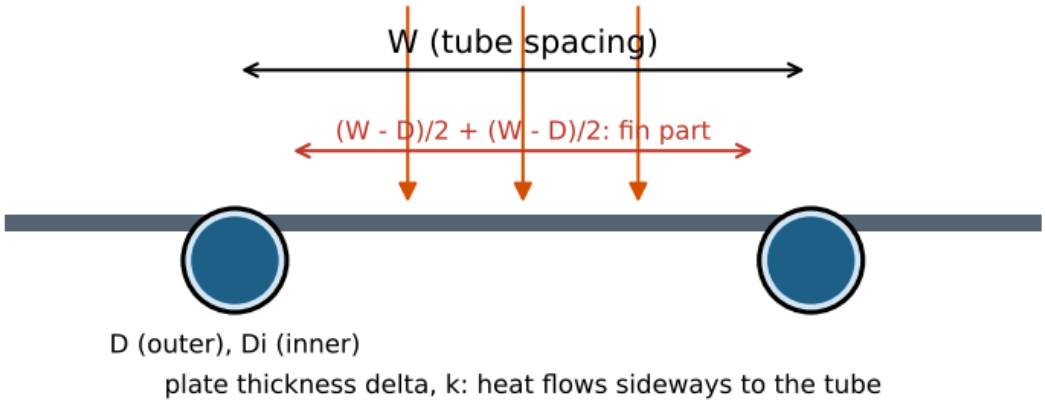
Thermal network for a double-glass-covered FPC (2021 5a)



Thermal network of a single-cover FPC (add one more stage for two covers)

For two covers the top path has three stages in series: plate T_p → $[h_{c,p\text{-}c1} \parallel h_{r,p\text{-}c1}]$ → cover 1 T_{c1} → $[h_{c,c1\text{-}c2} \parallel h_{r,c1\text{-}c2}]$ → cover 2 T_{c2} → $[h_w \parallel h_{r,c2\text{-}s}]$ → ambient/sky. The bottom path is plate → insulation conduction (L/k) → $[h_{c,b\text{-}a} \parallel h_{r,b\text{-}a}]$ → ambient, and edge losses in parallel. Absorbed solar S enters at the plate node; Q_u leaves to the fluid. So $U_t = [1/(h_c + h_r)_{p\text{-}c1} + 1/(h_c + h_r)_{c1\text{-}c2} + 1/(h_w + h_r)_{c2\text{-}a}]^{-1}$.

Derivation of the collector efficiency factor F′ (2022 5b)



Sheet-and-tube absorber: tube spacing W, tube diameter D

Step (write this)	Why / what it means
Fin of length (W - D)/2 on each side of a tube, thickness δ, conductivity k; absorbed S, loss $U_L(T - T_a)$	Each half-plate between tubes is a fin heated by the sun.
Fin equation: $d^2T/dx^2 = (U_L/k\delta)[T - T_a - S/U_L]$; $m^2 = U_L/(k\delta)$	Energy balance on a strip dx of the fin.
Heat reaching the tube from both fins: $q'_{fin} = (W - D)F[S - U_L(T_b - T_a)]$, $F = \tanh[m(W - D)/2]/[m(W - D)/2]$	Standard fin solution (T_b = base temperature at the tube).
Gain over the tube itself: $q'_{tube} = D[S - U_L(T_b - T_a)] \Rightarrow q'_u = [(W - D)F + D][S - U_L(T_b - T_a)]$	Total per unit length of tube.
Same heat crosses bond and tube wall to the fluid: $q'_u = (T_b - T_f)/[1/(h_{fi}\pi D_i) + 1/C_b]$	C_b = bond conductance.
Eliminate T_b between the two equations: $q'_u = WF'[S - U_L(T_f - T_a)]$	Algebra: solve the second for T_b and substitute.
$F' = (1/U_L) / \{\{W[1/(U_L(D + (W - D)F)) + 1/C_b + 1/(\pi D_i h_{fi})]\}\}$	Symbols: W tube spacing, D outer and D_i inner tube diameter, F fin efficiency, C_b bond conductance, h_{fi} fluid-side coefficient, U_L overall loss coefficient.

Testing a new flat plate collector (2022 6a)

1. Mount the collector outdoors (or in a solar simulator) facing the sun, near normal incidence around solar noon, on a clear day (ASHRAE 93 / ISO 9806).
2. Circulate fluid at a fixed flow rate (about 0.02 kg/s per m^2) with controlled inlet temperature; measure $\dot{m}$, T_i, T_o, irradiance I_T (pyranometer in the collector plane), T_a and wind speed.
3. Record quasi-steady periods (10-15 min) at at least four inlet temperatures from near ambient to high.
4. For each: $\eta = \dot{m}c_p(T_o - T_i)/(A_c I_T)$; plot η against $(T_i - T_a)/I_T$. The straight line gives intercept $F_R(\tau\alpha)_n$ and slope $-F_R U_L$ (Hottel-Whillier-Bliss).
5. Also measure the **time constant** (shade and unshade, time to reach 63.2% of the change) and the **incidence angle modifier** $K_{\tau\alpha}$ at several angles; stagnation and pressure/leak tests for durability.

Numerical 1: top and overall loss coefficient with known glass temperature (2021 5c)

Note: the air gap is printed as "75 cm", impossible for a collector; it is taken as **7.5 cm (75 mm)**. Wind speed is not given; it is not needed because the glass temperature is known.

Step (write this)	Why / what it means
Mean gap air temperature (70 + 50)/2 = 60 °C: k = 0.0288 W/m K, $\nu = 1.897\times10^{-5}$, $\alpha = 2.697\times10^{-5}$ m^2/s, $\beta' = 1/333\ K^{-1}$	Air table.
$Ra = g\beta'(T_p - T_g)L^3/(\nu\alpha) = 9.81 \times 20 \times 0.075^3/(333 \times 1.897\times10^{-5} \times 2.697\times10^{-5}) = 4.86\times10^5$; $Ra\cos45° = 3.43\times10^5$	
Hollands: $Nu = 1 + 1.44[1 - 1708/(Ra\cos\beta)][1 - 1708(\sin81°)^{1.6}/(Ra\cos\beta)] + [(Ra\cos\beta/5830)^{1/3} - 1] = 5.32$	Tiwari Eq. 5.17b (β = 45°).
$h_{c,p\text{-}g} = Nu\ k/L = 5.32 \times 0.0288/0.075 = 2.04\ W/m^2K$	
$\varepsilon_{eff} = 1/(1/0.95 + 1/0.8 - 1) = 0.768$; $h_{r,p\text{-}g} = \varepsilon_{eff}\sigma(T_p^2 + T_g^2)(T_p + T_g) = 0.768 \times 5.67\times10^{-8} \times (343.15^2 + 323.15^2)(666.3) = 6.44\ W/m^2K$	Linearised radiation coefficient (K).
Top loss flux $q_t = (h_c + h_r)(T_p - T_g) = 8.49 \times 20 = 169.7\ W/m^2$	Everything that crosses the gap reaches the ambient (steady state).
$U_t = q_t/(T_p - T_a) = 169.7/50 = 3.39\ W/m^2K$; $U_b = k/L = 0.03/0.05 = 0.60\ W/m^2K$	Edge loss neglected (no edge data).
$U_L = U_t + U_b = 3.99 \approx 4.0\ W/m^2K$	

Numerical 2: overall loss coefficient by iteration (2019 5b)

$$Nu = 1 + 1.44\left[1 - \frac{1708}{Ra\cos\beta}\right]^{+}\left(1 - \frac{1708[\sin(1.8)\beta]^{1.6}}{Ra\cos\beta}\right) + \left\{\left[\frac{Ra\cos\beta}{5830}\right]^{1/3} - 1\right\}^{+}$$

Given correlations (2019 paper)

Step (write this)	Why / what it means
$h_w = 8.6V^{0.6}/L^{0.4} = 8.6 \times 2.5^{0.6}/8^{0.4} = 6.49\ W/m^2K$; $\varepsilon_{eff} = 1/(1/0.95 + 1/0.88 - 1) = 0.841$; $T_{sky} = T_a - 6 = 14\ °C$	Wind formula as given; sky temperature as in Tiwari and Hasan sir's notes.
Guess $T_g = 40\ °C$: gap air at 54.5 °C, $Ra = 7.6\times10^5$, $Nu = 5.95$, $h_{c,p\text{-}g} = 2.25$, $h_{r,p\text{-}g} = 6.72$, $h_{r,g\text{-}a} = \varepsilon_g\sigma(T_g^4 - T_{sky}^4)/(T_g - T_a) = 7.03$	Step I of the iteration.
$U_t = [1/(2.25 + 6.72) + 1/(6.49 + 7.03)]^{-1} = 5.39\ W/m^2K$	Two stages in series.
New $T_g = T_p - U_t(T_p - T_a)/(h_c + h_r)_{p\text{-}g} = 69 - 5.39 \times 49/8.97 = 39.55\ °C$	Check the guess; repeat.

Converged: T_g = 39.5 °C, Nu = 5.98, $h_{c,p\text{-}g}$ = 2.26, $h_{r,p\text{-}g}$ = 6.71, $h_{r,g\text{-}a}$ = 7.05 ⇒ **U_t = 5.40 W/m²K**	One more pass changes nothing.
$U_b = k/L_b = 0.04/0.06 = 0.667$ W/m²K	
$U_e = (k/L_e)$(edge area/A_c) = (0.04/0.02) × [2(8 + 2.5) × 0.08]/(8 × 2.5) = 2 × 1.68/20 = 0.168 W/m²K	Edge area = perimeter × collector thickness.
U_L = 5.40 + 0.667 + 0.168 = 6.23 W/m²K	If the sky is taken at ambient temperature (Kalogirou), U_t = 5.15 and U_L = 5.99 W/m²K.

Numerical 3: fin efficiency, F′, useful energy, efficiency (2020 4c)

Step (write this)	Why / what it means
$m = \sqrt{(U_L/k\delta)} = \sqrt{(6.9/(385 \times 0.0004))} = 6.69$ m^{-1}; $m(W - D)/2 = 6.69 \times 0.105/2 = 0.351$	
(i) F = tanh(0.351)/0.351 = 0.961	Fin efficiency.
$1/(U_L[D + (W - D)F]) = 1/(6.9 \times [0.015 + 0.105 \times 0.961]) = 1.255$; $1/(\pi D_i h_{fi}) = 1/(\pi \times 0.0135 \times 320) = 0.0737$	Bond resistance ignored ($1/C_b$ = 0).
(ii) F′ = (1/6.9)/[0.12 × (1.255 + 0.0737)] = 0.1449/0.1594 = 0.912	
$F'' = (\dot{m}c_p/A_cU_LF')[1 - \exp(-A_cU_LF'/\dot{m}c_p)]$ with $\dot{m}c_p$ = 0.06 × 4180 = 250.8 W/K: = $9.97[1 - e^{-0.1003}]$ = 0.951; $F_R = F'F''$ = 0.868	Flow factor.
$I_T = 2.88\times10^6/3600$ = 800 W/m²; $S = (\tau\alpha)I_T$ = 0.8 × 800 = 640 W/m²	Hourly irradiation → average irradiance.
(iii) $Q_u = A_cF_R[S - U_L(T_i - T_a)]$ = 4 × 0.868 × (640 - 6.9 × 5) = 2102 W (7.57 MJ in the hour)	"Temperature difference 5 °C" read as $T_i - T_a$.
(iv) $n = Q_u/(A_cI_T)$ = 2102/(4 × 800) = 65.7%	

Numerical 4: U_L from a dark test (2022 5c)

Step (write this)	Why / what it means
$\dot{m}$ = 0.05 L/s × 0.980 kg/L = 0.0490 kg/s (water at 65 °C)	
Heat lost $Q = \dot{m}c_p(T_i - T_o)$ = 0.0490 × 4180 × 15 = 3074 W	No sun: the collector only loses heat; the fluid cools.
Mean fluid temperature T_m = (65 + 50)/2 = 57.5 °C; $Q = A_cU_L(T_m - T_a)$	Plate assumed close to the mean fluid temperature.
U_L = 3074/(5 × (57.5 - 8)) = 12.4 W/m²K	Exact exponential form: $\ln[(T_i - T_a)/(T_o - T_a)] = A_cF'U_L/\dot{m}c_p$ gives $F'U_L$ = 12.5: same. (With ρ = 1000 kg/m³, 12.7.) High value: probably an unglazed or poorly insulated test collector.

Memory trick: "F′ local fluid, F_R inlet fluid". Efficiency line: intercept $F_R(\tau\alpha)$, slope $-F_RU_L$. U_t = two parallel pairs in series; U_b = k/L; $U_e = U_{b,edge}A_e/A_c$.